Supplementary Material

Reliability and Deployment Challenges of Visual Foundation Models in Visual Computing: A Task-Centric Survey

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Reproducible Resource Package — Version 1.0

Contents

Supplementary Material — Overview and Purpose

This document is the **reproducible supplementary resource** accompanying the survey *"Reliability and Deployment Challenges of Visual Foundation Models in Visual Computing: A Task-Centric Survey."* It is intended to be released as an **open resource** through a public repository (GitHub) with an archived, citable snapshot (Zenodo) carrying a permanent Digital Object Identifier (DOI).

Although the main manuscript is a **review article** rather than an experimental paper, this supplement follows the same open-science principles recommended for non-review submissions. It packages, in fully documented and machine-readable form, every structured artefact underlying the survey:

1. The complete **literature corpus spreadsheet** (all surveyed papers, with metadata, category labels, and the venue/year used in the methodology section).
2. The full set of **taxonomy labels** assigning every surveyed model to a category under the decision rules of the main paper.
3. All **benchmark tables** in machine-readable form, with explicit provenance flags.
4. The complete **weakness annotation set** (the 19 documented weaknesses), including the per-factor severity scores, evidence types, affected applications, and remediation pathways.
5. Supporting material: the severity-scoring rubric as executable pseudo-code, the deployment-threshold table, a data dictionary, a reproducibility checklist, and licensing/citation information.

The goal is to allow any reader to **inspect, verify, re-use, and extend** the survey's structured claims — for example, to recompute severity labels, to filter the corpus by venue or year, or to add new models to the taxonomy using the documented rules.

1. Repository Structure and Access

1.1 Proposed Repository Layout

The open resource is organised as a single repository with the following recommended structure. (Placeholder paths are shown; the authors will finalise the URL and DOI upon acceptance.)

```
vfm-visual-computing-survey/
|-- README.md                     # Landing page, citation, license
|-- LICENSE                       # CC BY 4.0 (data) + MIT (code)
|-- CITATION.cff                  # Machine-readable citation metadata
|-- CHANGELOG.md                  # Versioned change history
|
|-- data/
|   |-- literature_corpus.csv     # Full surveyed-paper spreadsheet
|   |-- literature_corpus.xlsx    # Same, spreadsheet form
|   |-- taxonomy_labels.csv       # Model → category assignments
|   |-- weakness_annotations.csv  # 19 weaknesses, all fields
|   |-- benchmarks/
|   |   |-- generative_models.csv
|   |   |-- discriminative_models.csv
|   |   |-- neural_rendering.csv
|   |   |-- depth_estimation.csv
|   |   |-- image_enhancement.csv
|   |   |-- medical_vfms.csv
|   |   |-- clip_variants.csv
|   |   |-- sam_family.csv
|   |   |-- nerf_3dgs.csv
|   |   `--- peft_strategies.csv
|   `--- deployment_thresholds.csv
|
|-- schema/
|   |-- data_dictionary.md        # Field-by-field definitions
|   `--- severity_rubric.py        # Executable severity calculator
|
|-- docs/
|   |-- methodology.md            # PRISMA process, search strings
|   |-- reproducibility.md        # How to reproduce every table/figure
|   `--- taxonomy_decision_rules.md # R1-R3 with worked examples
|
`--- scripts/
    |-- validate_data.py          # Schema/consistency checks
    |-- recompute_severity.py     # Reproduces severity labels
    `--- corpus_stats.py           # Regenerates publication-trend stats
```

1.2 Access, Permanence, and DOI

- **Development repository (GitHub):** https://github.com/<org>/vfm-visual-computing-survey *(to be finalised on acceptance)*

- **Archived snapshot (Zenodo):** each tagged release is mirrored to Zenodo, which mints a version-specific DOI plus a "concept" DOI that always resolves to the latest version.
- **Permanent citation DOI:** `https://doi.org/10.5281/zenodo.XXXXXXX` *(reserved; to be activated on publication)*
- **License:** data files under **Creative Commons Attribution 4.0 (CC BY 4.0)**; helper scripts under the **MIT License**. Both permit reuse with attribution.

1.3 Citation Note

Users of this resource are asked to cite both the article and the archived dataset:

> Madaan, N., Malik, R., Kumar, A., Upadhaya, U. *Reliability and Deployment Challenges of Visual Foundation Models in Visual Computing: A Task-Centric Survey.* The Visual Computer (2025). Accompanying reproducible resource: Zenodo, DOI: 10.5281/zenodo.XXXXXXX.

A ready-to-use `CITATION.cff` file is included in the repository so that GitHub and Zenodo can render the citation automatically, and so reference managers can import it directly.

2. Data Dictionary

Every column used across the supplementary data files is defined here once, so that the CSV/XLSX files are self-describing. Fields are grouped by file.

2.1 `literature_corpus.csv`

Field	Type	Description
`ref_id`	string	Internal citation key (matches the manuscript `.bib`).
`authors`	string	Author list (surname, initials).
`title`	string	Full paper title.
`venue`	string	Publication venue (journal or conference).
`year`	integer	Publication year.
`venue_type`	enum	`journal` / `conference` / `preprint`.
`category`	enum	Taxonomy category: `Generative` / `Discriminative` / `Hybrid`.
`model_family`	string	Model family if the paper introduces one (e.g., CLIP, SAM).
`vc_area`	string	Primary visual-computing area (rendering, 3D, VLM, ...).
`peer_reviewed`	boolean	`TRUE` if peer-reviewed at time of writing.
`screening_stage`	enum	`included` (all corpus rows are `included`).
`notes`	string	Free-text annotation.

2.2 `taxonomy_labels.csv`

Field	Type	Description
`model`	string	Model name.
`primary_category`	enum	`Generative` / `Discriminative` / `Hybrid`.
`primary_output`	string	The artefact the model produces (rule R1).
`rule_applied`	string	Which rule(s) decided the label (R1/R2/R3).
`overlap_notes`	string	Secondary roles / cross-category behaviour.

2.3 `weakness_annotations.csv`

Field	Type	Description
`weakness_id`	string	`W1`–`W19`.
`model_family`	string	Affected model family.
`description`	string	One-line description of the weakness.
`evidence_type`	enum	`E1` empirical / `E2` multi-study / `E3` analytical.
`impact`	integer (1–3)	Severity factor I.
`breadth`	integer (1–3)	Severity factor B.
`reversibility`	integer (1–3)	Severity factor R.
`score_S`	integer (3–9)	Composite `I+B+R`.
`severity`	enum	`H` ($S \geq 7$) / `M` ($S \leq 6$).
`affected_applications`	string	VC domains materially affected.
`remediation`	string	Mitigation strategy / representative method.

2.4 `benchmarks/*.csv`

Each benchmark file shares a common provenance convention:

Field	Type	Description
`model`	string	Model / method name.
`year`	integer	Year.
`metric_*`	number	Reported metric (column name states the metric & dataset).
`setting`	enum	`ZS` zero-shot / `LP` linear-probe / `FT` fine-tuned.
`provenance`	enum	`original` (†) / `secondary` (‡) / `approx` (≈) / `non_pr` (∗).
`source_ref`	string	`ref_id` of the source paper.

2.5 `deployment_thresholds.csv`

Field	Type	Description
`scenario`	string	Deployment scenario (AR/VR, edge, medical, . . .).
`threshold`	string	Concrete quantitative target.
`violated_by`	string	Weakness IDs that violate the threshold.
`current_gap`	string	Description of the gap.

3. Survey Methodology (Extended)

This section expands the methodology of the main paper into a fully reproducible protocol.

3.1 Search Strings (verbatim)

The following Boolean strings were issued to each database. Dates and result counts are recorded in `docs/methodology.md` in the repository.

Model-term string:

```
("foundation model" OR "vision-language model" OR "CLIP" OR
 "segment anything" OR "SAM" OR "diffusion model" OR "DINO" OR
 "neural radiance field" OR "NeRF" OR "Gaussian splatting")
```

Task-term string:

```
("neural rendering" OR "3D reconstruction" OR "image synthesis" OR
 "segmentation" OR "depth estimation" OR "visual question answering" OR
 "medical image" OR "augmented reality")
```

Analysis-term string:

```
("survey" OR "benchmark" OR "robustness" OR "limitation" OR
 "weakness" OR "hallucination" OR "deployment")
```

Composite query (representative):

```
("foundation model" OR "vision-language model") AND
("visual computing" OR "neural rendering" OR "segmentation") AND
("robustness" OR "limitation" OR "benchmark")
```

3.2 Databases and Date of Search

Database	Role	Records identified
IEEE Xplore	primary	312
ACM Digital Library	primary	248
SpringerLink	primary	196
Scopus	primary	274
arXiv	preprints	358
Semantic Scholar	primary	92
Citation snowballing	supplementary	95
Total (pre-dedup)		**1,575**

3.3 PRISMA Flow (counts)

```
Identification:  1,575 records  (1,480 database + 95 snowball)
   |  remove duplicates (−412)
Screening:       1,163 records screened on title/abstract
   |  exclude off-topic / out of scope (−821)
Eligibility:       342 full-text articles assessed
```

```
   |  exclude (−216): insufficient detail (E3-fail),
   |                  unverifiable (E4), superseded (E2)
Included:          126 primary studies
```

3.4 Inclusion / Exclusion Criteria (reference)

- **I1** VFM or its application to a visual-computing task.
- **I2** English language.
- **I3** Peer-reviewed, or preprint with demonstrable impact.
- **I4** Methodological contribution, empirical evaluation, or documented limitation relevant to reliability/deployment.
- **E1** Unrelated to visual computing or foundation models.
- **E2** Duplicate / superseded / extended abstract.
- **E3** Non-archival note lacking methodological detail.
- **E4** Inaccessible full text or unverifiable results.

4. Literature Corpus Spreadsheet

The complete corpus is provided as `data/literature_corpus.csv` (and `.xlsx`). Below is the **full tabulation** of the surveyed model families and key references, with the category labels and venue metadata used in the survey. (For space, multi-author lists are abbreviated; the CSV carries the full lists.)

ref_id	Title	Authors	Venue	Year	Type	Category	VC area	Peer-rev.
`radford2021clip`	CLIP	Radford et al.	ICML	2021	conference	Discriminative	Vision-language	Yes
`kirillov2023sam`	Segment Anything (SAM)	Kirillov et al.	ICCV	2023	conference	Discriminative	Segmentation	Yes
`ravi2024sam2`	SAM 2	Ravi et al.	arXiv	2024	preprint	Discriminative	Segmentation	No
`oquab2023dinov2`	DINOv2	Oquab et al.	TMLR	2023	journal	Discriminative	Self-supervised	Yes
`rombach2022latentdiffusion`	Stable Diffusion (LDM)	Rombach et al.	CVPR	2022	conference	Generative	Generation	Yes
`ramesh2022dalle2`	DALL-E 2	Ramesh et al.	arXiv	2022	preprint	Generative	Generation	No
`mildenhall2021nerf`	NeRF	Mildenhall et al.	ECCV/CACM	2021	conference	Discriminative	Neural rendering	Yes
`kerbl20233dgs`	3D Gaussian Splatting	Kerbl et al.	ACM TOG	2023	journal	Discriminative	Neural rendering	Yes
`liu2023llava`	LLaVA	Liu et al.	NeurIPS	2023	conference	Hybrid	Vision-language	Yes
`openai2023gpt4`	GPT-4V	OpenAI	arXiv	2023	preprint	Hybrid	Vision-language	No
`dosovitskiy2021vit`	Vision Transformer (ViT)	Dosovitskiy et al.	ICLR	2021	conference	Discriminative	Backbone	Yes
`vaswani2017transformer`	Transformer	Vaswani et al.	NeurIPS	2017	conference	Discriminative	Backbone	Yes
`poole2022dreamfusion`	DreamFusion	Poole et al.	ICLR	2023	conference	Generative	Text-to-3D	Yes
`liu2023zero123`	Zero-1-to-3	Liu et al.	ICCV	2023	conference	Generative	Image-to-3D	Yes

ref_id	Title	Authors	Venue	Year	Type	Category	VC area	Peer-rev.
ma2024medsam	MedSAM	Ma & Wang	Nature Comm.	2024	journal	Discriminative	Medical seg.	Yes
li2023llavamed	LLaVA-Med	Li et al.	NeurIPS	2023	conference	Hybrid	Medical VQA	Yes
zhang2024biomedclip	BiomedCLIP	Zhang et al.	NEJM AI	2024	journal	Discriminative	Medical VLM	Yes
liu2024unification	VFM unification survey	Liu et al.	The Visual Computer	2024	journal	Discriminative	Survey	Yes
yang2026vlmar	VLFM for AR/VR survey	Yang et al.	The Visual Computer	2025	journal	Hybrid	Survey/AR-VR	Yes
senior2025gnnvl	GNN in VL understanding	Senior et al.	The Visual Computer	2025	journal	Hybrid	VL understanding	Yes
bian2025mvsregnerf	MVSRegNeRF	Bian et al.	The Visual Computer	2025	journal	Discriminative	Neural rendering	Yes
clipgaussian2024	CLIPGaussian	Liu et al.	The Visual Computer	2025	journal	Discriminative	Rendering/restoration	Yes
yuan2025vcgs	VC-GS	Yuan et al.	The Visual Computer	2025	journal	Discriminative	Rendering/restoration	Yes
dtgs2025	DTGS	Dai & Yao	Comp. Anim. Virtual Worlds	2025	journal	Discriminative	View synthesis	Yes
agent2025hci	LLM agent interaction	Liu, Y.	VR & Intell. Hardware	2025	journal	Hybrid	HCI/agents	Yes

Note. The table above lists the principal model-introducing and survey references for readability. The machine-readable `literature_corpus.csv` contains **all 126 included studies** with complete author lists, DOIs, and the screening provenance for each entry.

5. Taxonomy Labels

This section provides the complete model-to-category assignment used in the survey, produced by applying the decision rules of the main paper:

- **R1 (Primary output):** synthesised content → Generative; structured interpretation → Discriminative.
- **R2 (Unified interface):** interleaves generation + understanding through a shared interface → Hybrid/Unified (precedence over R1).
- **R3 (Dominant-use tie-break):** a model reused as a component is classified by its own standalone output; secondary roles recorded as overlaps.

Model	Category	Primary output	Rule	Overlap notes
CLIP	Discriminative	Aligned image/text embeddings	R1, R3	Central overlap: reused in generative + hybrid pipelines
DINOv2	Discriminative	Dense self-supervised features	R1	Backbone reused widely
SAM / SAM2	Discriminative	Segmentation masks	R1	Promptable; used in editing pipelines
ViT	Discriminative	Patch embeddings / class logits	R1	Architectural backbone
NeRF	Discriminative	Per-scene 3D reconstruction (radiance field)	R1, R3	Straddles gen–disc (synthesises views)
3D Gaussian Splatting	Discriminative	Per-scene 3D reconstruction (Gaussians)	R1, R3	Straddles gen–disc
Stable Diffusion	Generative	Synthesised image	R1	Uses CLIP text encoder internally
DALL-E 2	Generative	Synthesised image	R1	—
Sora	Generative	Synthesised video	R1	—
DreamFusion	Generative	Synthesised 3D asset	R1	Uses 2D diffusion prior (SDS)
Zero-1-to-3	Generative	Novel-view image / 3D	R1	—
LLaVA	Hybrid	Interleaved text + visual response	R2	Frozen CLIP encoder + LLM
GPT-4V	Hybrid	Interleaved text + visual response	R2	—

Model	Category	Primary output	Rule	Overlap notes
BLIP-2	Hybrid	Interleaved text + visual response	R2	Q-Former bridges encoder + LLM
LLaVA-Med	Hybrid	Medical VQA response	R2	Domain-adapted LLaVA
BiomedCLIP	Discriminative	Medical image/text embeddings	R1	Medical CLIP variant

6. Weakness Annotations (Full Detail)

This is the core reproducible artefact of the survey: the complete set of 19 documented weaknesses, each expanded into a structured entry. The machine-readable form is `data/weakness_annotations.csv`. Every entry follows the four-dimension schema (evidence type, affected applications, severity, remediation) and reports the per-factor severity scores so the label can be recomputed from Equation (1) of the main paper:

> $\mathbf{S} = \mathbf{Impact\ (I)} + \mathbf{Breadth\ (B)} + \mathbf{Reversibility\ (R)}$, with **High if** $\mathbf{S \geq 7}$, **Medium if** $\mathbf{S \leq 6}$.

Scoring scales (1–3): Impact — 1 minor, 2 substantial, 3 task failure/unsafe; Breadth — 1 single task, 2 one domain, 3 multiple domains; Reversibility — 1 robust workaround, 2 partial, 3 none.

W1 — Fine-grained detail blindness

- **Model family:** CLIP
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=2, Breadth B=3, Reversibility R=2 → **S = 7** → **High (H)**
- **Affected applications:** Rendering, 3D generation, content creation
- **Root cause / mechanism:** CLIP's global contrastive objective optimises image-level alignment, so fine-grained attributes (small parts, textures, counts) are under-represented in the embedding.
- **Remediation pathway:** Region-aware or dense CLIP variants (e.g., FG-CLIP); patch-level contrastive objectives; feature pyramids.

W2 — Spatial reasoning near-random

- **Model family:** CLIP
- **Evidence type:** E2 (reported by multiple independent studies)
- **Severity score:** Impact I=3, Breadth B=3, Reversibility R=2 → **S = 8** → **High (H)**
- **Affected applications:** 3D generation, AR/VR, robotics
- **Root cause / mechanism:** Relational concepts (left/right/above/below) are handled near-randomly; the contrastive objective provides no explicit positional supervision.
- **Remediation pathway:** Spatially-aware pre-training; position-sensitive heads; structured scene-graph supervision.

W3 — Center bias (off-center objects lost)

- **Model family:** CLIP
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=2, Breadth B=2, Reversibility R=2 → **S = 6** → **Medium (M)**
- **Affected applications:** Image retrieval, content editing
- **Root cause / mechanism:** Off-center objects can vanish from the final embedding due to information loss during visual token pooling.
- **Remediation pathway:** Multi-crop / sliding-window aggregation; attention pooling that preserves peripheral tokens.

W4 — Effective text limited to ~20 tokens

- **Model family:** CLIP
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=2, Breadth B=3, Reversibility R=2 → **S = 7** → **High (H)**
- **Affected applications:** AI content creation, scene generation
- **Root cause / mechanism:** Despite a 77-token context, effective text understanding saturates near ~20 tokens, truncating long prompts.
- **Remediation pathway:** Long-context text encoders; prompt compression; hierarchical caption encoding.

W5 — Medical imaging failure (zero-shot Dice 58.5%)

- **Model family:** SAM/SAM2
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=3, Breadth B=2, Reversibility R=2 → **S = 7** → **High (H)**
- **Affected applications:** Medical visual computing
- **Root cause / mechanism:** Trained on natural images (SA-1B), SAM transfers poorly to medical modalities; mean zero-shot Dice 58.5% vs >90% for specialised models.
- **Remediation pathway:** Domain fine-tuning (MedSAM, 1.5M masks); medical adapters; modality-specific prompting.

W6 — Camouflaged / low-contrast objects fail

- **Model family:** SAM/SAM2
- **Evidence type:** E2 (reported by multiple independent studies)
- **Severity score:** Impact I=2, Breadth B=3, Reversibility R=2 → **S = 7** → **High (H)**
- **Affected applications:** Industrial inspection, remote sensing
- **Root cause / mechanism:** Objects with low figure-ground contrast are missed; SAM relies on appearance cues that fail under camouflage.
- **Remediation pathway:** Contrastive boundary refinement; frequency-domain cues; task-specific fine-tuning.

W7 — Prompt-dependent (not fully automatic)

- **Model family:** SAM/SAM2
- **Evidence type:** E2 (reported by multiple independent studies)
- **Severity score:** Impact I=2, Breadth B=2, Reversibility R=2 → **S = 6** → **Medium (M)**
- **Affected applications:** Automated visual pipelines
- **Root cause / mechanism:** SAM requires point/box prompts; fully unattended segmentation is unreliable, limiting batch automation.
- **Remediation pathway:** Automatic prompt generation; integration with detectors; everything-mode post-filtering.

W8 — Thin / elongated boundaries missed

- **Model family:** SAM/SAM2
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=2, Breadth B=2, Reversibility R=2 → **S = 6** → **Medium (M)**

- **Affected applications:** Digital humans, vessel segmentation
- **Root cause / mechanism:** Thin structures (hair, vessels, wires) are under-segmented due to mask-decoder resolution limits.
- **Remediation pathway:** High-resolution decoders; boundary-aware losses; multi-scale refinement.

W9 — Per-scene optimisation (<1 FPS)

- **Model family:** NeRF
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=3, Breadth B=3, Reversibility R=1 → **S = 7** → **High (H)**
- **Affected applications:** Real-time rendering, surgical AR
- **Root cause / mechanism:** Vanilla NeRF requires per-scene optimisation and renders at <0.03 FPS, incompatible with real-time/interactive budgets.
- **Remediation pathway:** Feed-forward / generalisable NeRF; 3DGS rasterisation; baked representations; distillation.

W10 — Sparse-viewpoint failure

- **Model family:** NeRF
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=3, Breadth B=2, Reversibility R=2 → **S = 7** → **High (H)**
- **Affected applications:** 3D reconstruction, heritage digitisation
- **Root cause / mechanism:** With few input views, geometry is under-constrained, producing floaters and incorrect surfaces.
- **Remediation pathway:** Depth/MVS regularisation (e.g., MVSRegNeRF); diffusion priors; few-shot NeRF.

W11 — Dynamic scenes unsupported (vanilla)

- **Model family:** NeRF
- **Evidence type:** E3 (analytical (architectural/objective consequence))
- **Severity score:** Impact I=2, Breadth B=2, Reversibility R=2 → **S = 6** → **Medium (M)**
- **Affected applications:** Avatars, video, surgical scenes
- **Root cause / mechanism:** The static radiance-field assumption breaks for moving content; vanilla NeRF cannot model temporal change.
- **Remediation pathway:** Dynamic / 4D NeRF; deformation fields; time-conditioned representations.

W12 — Janus / multi-view 3D inconsistency

- **Model family:** Stable Diffusion
- **Evidence type:** E2 (reported by multiple independent studies)
- **Severity score:** Impact I=3, Breadth B=2, Reversibility R=2 → **S = 7** → **High (H)**
- **Affected applications:** 3D asset generation, VFX, digital twins
- **Root cause / mechanism:** 2D-diffusion-guided 3D generation produces multi-face ('Janus') artefacts due to view-agnostic priors.
- **Remediation pathway:** Multi-view diffusion; 3D-aware score distillation; view-consistent conditioning.

W13 — Coarse semantic alignment

- **Model family:** Stable Diffusion
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=2, Breadth B=3, Reversibility R=2 → **S = 7** → **High (H)**
- **Affected applications:** AI content creation
- **Root cause / mechanism:** Generated content often misses fine prompt semantics (attributes, relations, counts).
- **Remediation pathway:** Stronger text encoders; attention guidance; compositional conditioning.

W14 — Social bias amplification

- **Model family:** Stable Diffusion
- **Evidence type:** E2 (reported by multiple independent studies)
- **Severity score:** Impact I=2, Breadth B=2, Reversibility R=2 → **S = 6** → **Medium (M)**
- **Affected applications:** Medical, educational, heritage
- **Root cause / mechanism:** Training-data biases are amplified in generated imagery (occupation, demographic stereotypes).
- **Remediation pathway:** Bias-aware fine-tuning; balanced data; fairness-constrained sampling.

W15 — Poor 3D awareness

- **Model family:** DINOv2
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=3, Breadth B=3, Reversibility R=2 → **S = 8** → **High (H)**
- **Affected applications:** 3D reconstruction, NeRF integration
- **Root cause / mechanism:** Strong 2D features lack 3D geometric consistency (probed by Probe3D), limiting use in 3D pipelines.
- **Remediation pathway:** 3D-aware pre-training; multi-view consistency objectives; geometry-grounded fine-tuning.

W16 — No native text understanding

- **Model family:** DINOv2
- **Evidence type:** E3 (analytical (architectural/objective consequence))
- **Severity score:** Impact I=1, Breadth B=3, Reversibility R=2 → **S = 6** → **Medium (M)**
- **Affected applications:** Text-guided VC, AI content tools
- **Root cause / mechanism:** DINOv2 is image-only; it cannot accept text prompts, blocking language-guided tasks without an added bridge.
- **Remediation pathway:** Align with a text encoder; distil into a VLM; CLIP-style post-hoc alignment.

W17 — 3D spatial reasoning failure

- **Model family:** VLMs
- **Evidence type:** E2 (reported by multiple independent studies)
- **Severity score:** Impact I=3, Breadth B=3, Reversibility R=2 → **S = 8** → **High (H)**

- **Affected applications:** AR/VR, 3D editing, robotics
- **Root cause / mechanism:** Multimodal VLMs struggle with 3D spatial relations and metric reasoning from 2D inputs.
- **Remediation pathway:** 3D-grounded instruction tuning; depth-augmented inputs; spatial probes.

W18 — Visual hallucination

- **Model family:** VLMs
- **Evidence type:** E2 (reported by multiple independent studies)
- **Severity score:** Impact I=3, Breadth B=3, Reversibility R=2 → **S = 8** → **High (H)**
- **Affected applications:** Medical, scientific visual computing
- **Root cause / mechanism:** VLMs generate fluent but unsupported visual claims (objects, measurements) absent from the image.
- **Remediation pathway:** RLHF grounding; POPE-style evaluation; retrieval-augmented grounding; abstention.

W19 — Real-time inference infeasible

- **Model family:** VLMs
- **Evidence type:** E1 (empirical / benchmarked)
- **Severity score:** Impact I=3, Breadth B=3, Reversibility R=1 → **S = 7** → **High (H)**
- **Affected applications:** AR/VR, robotics, interactive rendering
- **Root cause / mechanism:** Billion-parameter VLMs exceed interactive latency/memory budgets (e.g., LLaVA-7B ~37s mobile encoding).
- **Remediation pathway:** Distillation (LiteVLM); quantisation; adaptive computation (AdaVFM); edge-optimised variants.

7. Reproducible Severity Calculator

The severity label of every weakness is fully determined by its three factor scores. The repository ships the following executable reference implementation (`schema/severity_rubric.py`). Running it on `weakness_annotations.csv` reproduces the `severity` column exactly.

```python
import csv

def severity(impact, breadth, reversibility):
    \"\"\"Composite severity per Eq.(1): S = I + B + R; High if S>=7 else Medium.\"\"\"
    for name, v in [("impact", impact), ("breadth", breadth),
                    ("reversibility", reversibility)]:
        if v not in (1, 2, 3):
            raise ValueError(f"{name} must be in 1..3, got {v}")
    S = impact + breadth + reversibility
    return S, ("H" if S >= 7 else "M")

def verify(path="data/weakness_annotations.csv"):
    ok = True
    with open(path) as f:
        for row in csv.DictReader(f):
            S, label = severity(int(row["impact"]),
                                int(row["breadth"]),
                                int(row["reversibility"]))
            if S != int(row["score_S"]) or label != row["severity"]:
                print(f"MISMATCH at {row['weakness_id']}")
                ok = False
    print("All severity labels reproduced." if ok else "Discrepancies found.")
    return ok

if __name__ == "__main__":
    verify()
```

Verification result. Applying this calculator to all 19 weaknesses reproduces every published label (13 High, 6 Medium) with no discrepancies.

13. Extended Weakness Dossiers (Complete Set)

This section provides a self-contained dossier for **every** one of the 19 weaknesses. Each dossier uses an identical six-part structure — *Statement*, *Evidence and provenance*, *Visual-computing consequence*, *Severity derivation*, *Interaction with other weaknesses*, and *Remediation landscape* — so the catalogue can be read linearly or filtered programmatically against `data/weakness_annotations.csv`. All quantitative claims correspond to values recorded in that file and are reproduced from their original sources without normalisation.

Dossier W1: Fine-grained detail blindness

Model family: CLIP · **Evidence type:** E1 (empirical / benchmarked)

Statement. CLIP encodes each image into a single global embedding optimised for image-level alignment with a caption. Fine-grained attributes — small parts, subtle textures, exact counts, material cues — are under-represented because the contrastive objective rewards coarse semantic matching over dense attribute fidelity.

Visual-computing consequence. In text-to-image and text-to-3D pipelines that consume CLIP as a prior, fine-grained prompt terms are frequently ignored, reducing controllability and forcing manual post-editing of rendered or generated assets.

Severity derivation. Breadth=3 (propagates across every CLIP-conditioned generator); Impact=2 (degrades quality but the task still completes); Reversibility=2 (region-aware variants help but are not default). Composite $\mathbf{S = I{+}B{+}R = 2{+}3{+}2 = 7}$, mapped to **High (H)**.

Interaction with other weaknesses. Tightly coupled to W13 (coarse alignment in diffusion, which often reuses CLIP); compounds with W4 on long prompts.

Remediation landscape. Dense / region-aware CLIP; patch-level contrastive objectives; feature-pyramid conditioning; attribute-grounded fine-tuning.

ewpage ## Dossier W2: Spatial reasoning near-random

Model family: CLIP · **Evidence type:** E2 (reported by multiple independent studies)

Statement. Across standard spatial benchmarks, CLIP-based encoders handle relational concepts (left/right/above/below) at near-random accuracy; the global contrastive objective supplies no explicit positional supervision.

Visual-computing consequence. Compositional image synthesis, 3D scene layout, and AR object placement that rely on relational prompts are unreliable, because the encoder cannot distinguish spatial arrangements.

Severity derivation. Impact=3 (relational tasks effectively fail); Breadth=3 (3D gen, AR/VR, robotics); Reversibility=2 (spatial pre-training emerging). Composite $\mathbf{S = I{+}B{+}R = 3{+}3{+}2 = 8}$, mapped to **High (H)**.

Interaction with other weaknesses. Representation-level analogue of W17 (VLM spatial failure); compounds with W15 for 3D layout.

Remediation landscape. Spatially-aware pre-training; position-sensitive heads; scene-graph supervision; explicit layout conditioning.

ewpage ## Dossier W3: Center bias — off-center objects lost

Model family: CLIP · **Evidence type:** E1 (empirical / benchmarked)

Statement. Off-center objects can disappear from CLIP's final embedding because of information loss during visual-token pooling, biasing representation toward central content.

Visual-computing consequence. Retrieval and content-editing tools that depend on peripheral objects can miss them, producing incomplete edits or matches.

Severity derivation. Impact=2; Breadth=2 (retrieval and editing); Reversibility=2 (multi-crop aggregation mitigates). Composite $\mathbf{S} = \mathbf{I}+\mathbf{B}+\mathbf{R} = \mathbf{2}+\mathbf{2}+\mathbf{2} = \mathbf{6}$, mapped to **Medium (M)**.

Interaction with other weaknesses. Interacts with W1: both stem from pooling that discards spatially or semantically peripheral detail.

Remediation landscape. Multi-crop / sliding-window aggregation; attention pooling preserving peripheral tokens; higher input resolution.

ewpage ## Dossier W4: Effective text limited to ~20 tokens

Model family: CLIP · **Evidence type:** E1 (empirical / benchmarked)

Statement. Although the text branch accepts 77 tokens, effective utilisation saturates near ~20; later clauses contribute little to the embedding.

Visual-computing consequence. Long compositional prompts are silently truncated, dropping the often-specific later clauses that professional content creation relies on.

Severity derivation. Impact=2; Breadth=3 (all CLIP-conditioned generation); Reversibility=2. Composite $\mathbf{S} = \mathbf{I}+\mathbf{B}+\mathbf{R} = \mathbf{2}+\mathbf{3}+\mathbf{2} = \mathbf{7}$, mapped to **High (H)**.

Interaction with other weaknesses. Amplifies W1 and W13 for long descriptive prompts.

Remediation landscape. Long-context / hierarchical text encoders; prompt compression; segment-wise caption encoding.

ewpage ## Dossier W5: Medical-imaging segmentation failure

Model family: SAM/SAM2 · **Evidence type:** E1 (empirical / benchmarked)

Statement. Zero-shot SAM attains ~58.5% mean Dice on medical data versus >90% for specialised models; SA-1B (natural images) does not cover medical texture/contrast statistics.

Visual-computing consequence. Direct clinical use is precluded: ~0.58 Dice is well below the ~0.80 clinical-utility threshold, so SAM cannot be trusted for diagnostic segmentation unadapted.

Severity derivation. Impact=3 (clinical task fails); Breadth=2 (medical domain); Reversibility=2 (fine-tuning works but needs large curated data). Composite $\mathbf{S} = \mathbf{I}+\mathbf{B}+\mathbf{R} = \mathbf{3}+\mathbf{2}+\mathbf{2} = \mathbf{7}$, mapped to **High (H)**.

Interaction with other weaknesses. Shares the natural-image-pretraining root cause with the dataset-shift discussion; interacts with W8 for thin anatomy.

Remediation landscape. Domain fine-tuning (MedSAM, ~1.5M masks); medical adapters/LoRA; modality-specific prompting; uncertainty flags.

ewpage ## Dossier W6: Camouflaged / low-contrast objects fail

Model family: SAM/SAM2 · **Evidence type:** E2 (reported by multiple independent studies)

Statement. Objects with low figure-ground contrast are missed; SAM's appearance-driven cues fail under camouflage, reported across multiple studies.

Visual-computing consequence. Industrial inspection and remote sensing — where targets blend into background — see unreliable masks.

Severity derivation. Impact=2; Breadth=3 (multiple domains); Reversibility=2. Composite **S = I+B+R = 2+3+2 = 7**, mapped to **High (H)**.

Interaction with other weaknesses. Independent of medical W5 but shares the appearance-cue limitation.

Remediation landscape. Contrastive boundary refinement; frequency-domain cues; task-specific fine-tuning.

ewpage ## Dossier W7: Prompt-dependent (not fully automatic)

Model family: SAM/SAM2 · **Evidence type:** E2 (reported by multiple independent studies)

Statement. SAM needs point/box prompts; unattended segmentation is unreliable, limiting batch automation.

Visual-computing consequence. Fully automated visual pipelines (e.g., large-scale dataset labelling) cannot depend on SAM without a prompt generator.

Severity derivation. Impact=2; Breadth=2; Reversibility=2. Composite **S = I+B+R = 2+2+2 = 6**, mapped to **Medium (M)**.

Interaction with other weaknesses. Orthogonal to quality weaknesses; about autonomy rather than accuracy.

Remediation landscape. Automatic prompt generation; detector integration; everything-mode post-filtering.

ewpage ## Dossier W8: Thin / elongated boundaries missed

Model family: SAM/SAM2 · **Evidence type:** E1 (empirical / benchmarked)

Statement. Thin structures (hair, vessels, wires) are under-segmented due to mask-decoder resolution limits.

Visual-computing consequence. Digital-human grooming and medical vessel segmentation lose fine structure.

Severity derivation. Impact=2; Breadth=2; Reversibility=2. Composite **S = I+B+R = 2+2+2 = 6**, mapped to **Medium (M)**.

Interaction with other weaknesses. Compounds W5 in medical vessel tasks.

Remediation landscape. High-resolution decoders; boundary-aware losses; multi-scale refinement.

ewpage ## Dossier W9: Per-scene optimisation (<1 FPS)

Model family: NeRF · **Evidence type:** E1 (empirical / benchmarked)

Statement. Vanilla NeRF optimises one MLP per scene and renders at <0.03 FPS due to many per-ray evaluations.

Visual-computing consequence. Interactive/immersive uses (games, AR, surgical guidance) cannot use vanilla NeRF in the loop; per-scene cost blocks instant creation.

Severity derivation. Impact=3; Breadth=3; Reversibility=1 (3DGS/feed-forward provide robust workarounds) → S=7. Composite **S = I+B+R = 3+3+1 = 7**, mapped to **High (H)**.

Interaction with other weaknesses. Pairs with W19 as the two main real-time blockers.

Remediation landscape. 3DGS rasterisation (~140 FPS); Instant-NGP; feed-forward NeRF; baking/distillation.

ewpage ## Dossier W10: Sparse-viewpoint failure

Model family: NeRF · **Evidence type:** E1 (empirical / benchmarked)

Statement. With few input views geometry is under-constrained, yielding floaters and incorrect surfaces.

Visual-computing consequence. 3D reconstruction and heritage digitisation from limited captures produce unreliable geometry.

Severity derivation. Impact=3; Breadth=2; Reversibility=2. Composite **S = I+B+R = 3+2+2 = 7**, mapped to **High (H)**.

Interaction with other weaknesses. Mitigated by the same depth/MVS priors highlighted for MVSRegNeRF.

Remediation landscape. Depth/MVS regularisation; diffusion priors; few-shot NeRF.

ewpage ## Dossier W11: Dynamic scenes unsupported (vanilla)

Model family: NeRF · **Evidence type:** E3 (analytical (architectural or training-objective consequence))

Statement. The static radiance-field assumption breaks for moving content; vanilla NeRF cannot model temporal change.

Visual-computing consequence. Avatars, video, and surgical scenes with motion are not representable without extensions.

Severity derivation. Impact=2; Breadth=2; Reversibility=2 (4D NeRF exists). Composite **S = I+B+R = 2+2+2 = 6**, mapped to **Medium (M)**.

Interaction with other weaknesses. Independent of quality weaknesses; an expressivity limit.

Remediation landscape. Dynamic/4D NeRF; deformation fields; time-conditioned representations.

ewpage ## Dossier W12: Janus / multi-view 3D inconsistency

Model family: Stable Diffusion · **Evidence type:** E2 (reported by multiple independent studies)

Statement. 2D-diffusion-guided 3D generation produces multi-face ('Janus') artefacts because the prior is view-agnostic.

Visual-computing consequence. 3D asset generation, VFX, and digital twins receive geometrically inconsistent outputs needing manual repair.

Severity derivation. Impact=3; Breadth=2; Reversibility=2. Composite **S = I+B+R = 3+2+2 = 7**, mapped to **High (H)**.

Interaction with other weaknesses. Related to W15/W17 (3D awareness) at the generative side.

Remediation landscape. Multi-view diffusion; 3D-aware score distillation; view-consistent conditioning.

ewpage ## Dossier W13: Coarse semantic alignment

Model family: Stable Diffusion · **Evidence type:** E1 (empirical / benchmarked)

Statement. Generated content often misses fine prompt semantics — attributes, relations, counts.

Visual-computing consequence. AI content-creation controllability is limited; users iterate prompts manually.

Severity derivation. Impact=2; Breadth=3; Reversibility=2. Composite **S = I+B+R = 2+3+2 = 7**, mapped to **High (H)**.

Interaction with other weaknesses. Inherits W1/W4 via the CLIP text encoder used by many diffusion models.

Remediation landscape. Stronger text encoders; attention guidance; compositional conditioning.

ewpage ## Dossier W14: Social bias amplification

Model family: Stable Diffusion · **Evidence type:** E2 (reported by multiple independent studies)

Statement. Training-data biases are amplified in generated imagery (occupational, demographic stereotypes), reported across studies.

Visual-computing consequence. Medical, educational, and heritage content generation risk propagating stereotypes.

Severity derivation. Impact=2; Breadth=2; Reversibility=2. Composite **S = I+B+R = 2+2+2 = 6**, mapped to **Medium (M)**.

Interaction with other weaknesses. An ethics/robustness weakness distinct from quality weaknesses.

Remediation landscape. Bias-aware fine-tuning; balanced data; fairness-constrained sampling.

ewpage ## Dossier W15: Poor 3D awareness

Model family: DINOv2 · **Evidence type:** E1 (empirical / benchmarked)

Statement. Probing (Probe3D) shows strong 2D features carry limited 3D geometric information; multi-view consistency and metric depth are weakly encoded.

Visual-computing consequence. Using DINOv2 features to initialise/regularise 3D pipelines yields limited gains and can introduce geometric error.

Severity derivation. Impact=3; Breadth=3; Reversibility=2 → S=8. Composite **S = I+B+R = 3+3+2 = 8**, mapped to **High (H)**.

Interaction with other weaknesses. Representation-level sibling of W17; both reflect 2D pre-training bias.

Remediation landscape. 3D-aware pre-training; multi-view consistency objectives; geometry-grounded fine-tuning; depth fusion.

ewpage ## Dossier W16: No native text understanding

Model family: DINOv2 · **Evidence type:** E3 (analytical (architectural or training-objective consequence))

Statement. DINOv2 is image-only and cannot accept text prompts, blocking language-guided tasks without an added bridge.

Visual-computing consequence. Text-guided visual computing and AI content tools cannot use DINOv2 directly.

Severity derivation. Impact=1 (a capability gap, not an error); Breadth=3; Reversibility=2 → S=6. Composite **S = I+B+R = 1+3+2 = 6**, mapped to **Medium (M)**.

Interaction with other weaknesses. Complementary to CLIP weaknesses: DINOv2 lacks text, CLIP lacks dense detail.

Remediation landscape. Align with a text encoder; distil into a VLM; CLIP-style post-hoc alignment.

ewpage ## Dossier W17: 3D spatial reasoning failure

Model family: VLMs · **Evidence type:** E2 (reported by multiple independent studies)

Statement. Multimodal VLMs struggle with 3D spatial relations and metric reasoning from 2D inputs.

Visual-computing consequence. AR/VR, 3D editing, and robotics that need spatial grounding are unreliable.

Severity derivation. Impact=3; Breadth=3; Reversibility=2 → S=8. Composite **S = I+B+R = 3+3+2 = 8**, mapped to **High (H)**.

Interaction with other weaknesses. Behaviour-level counterpart of W15; co-occurs with W18.

Remediation landscape. 3D-grounded instruction tuning; depth-augmented inputs; spatial probes/benchmarks.

ewpage ## Dossier W18: Visual hallucination

Model family: VLMs · **Evidence type:** E2 (reported by multiple independent studies)

Statement. VLMs generate fluent but unsupported visual claims absent from the image, documented by multiple independent studies.

Visual-computing consequence. In clinical and scientific use a hallucinated finding is a safety event; fluency makes it hard to detect.

Severity derivation. Impact=3; Breadth=3; Reversibility=2 → S=8. Composite **S = I+B+R = 3+3+2 = 8**, mapped to **High (H)**.

Interaction with other weaknesses. Binding constraint for clinical/scientific deployment; co-occurs with W17.

Remediation landscape. RLHF grounding; POPE-style benchmarks; retrieval-augmented grounding; calibrated abstention.

ewpage ## Dossier W19: Real-time inference infeasible

Model family: VLMs · **Evidence type:** E1 (empirical / benchmarked)

Statement. Billion-parameter VLMs exceed interactive latency/memory budgets (e.g., ~37 s mobile encoding for a 7B model; >100 ms/query interactive).

Visual-computing consequence. Interactive AR/VR assistants, robotic loops, and in-the-loop tools cannot meet frame budgets with current VLMs.

Severity derivation. Impact=3; Breadth=3; Reversibility=1 → S=7. Composite **S = I+B+R = 3+3+1 = 7**, mapped to **High (H)**.

Interaction with other weaknesses. Pairs with W9 as the two real-time blockers; constrains the edge-memory threshold.

Remediation landscape. Distillation (LiteVLM); quantisation; adaptive computation (AdaVFM); edge-optimised variants; speculative decoding.

ewpage # 14. Per-Model Deep-Dive Appendices

These appendices collect, per model family, the extended variant tables that support the summary tables in the main paper. They are released as the corresponding CSVs under `data/benchmarks/`.

14.1 CLIP Family Variants

Variant	Architecture	Params	Res.	Train data	IN-1K ZS↑	Provenance
CLIP RN50	ResNet-50	~102M	224	WIT-400M	59.6	original
CLIP ViT-B/32	ViT-B	~150M	224	WIT-400M	63.3	original
CLIP ViT-B/16	ViT-B	~150M	224	WIT-400M	68.3	original
CLIP ViT-L/14	ViT-L	~428M	224	WIT-400M	75.5	original
CLIP ViT-L/14@336	ViT-L	~428M	336	WIT-400M	76.6	original
OpenCLIP ViT-G/14	ViT-G	~1.8B	224	LAION-2B	80.1	secondary
SigLIP ViT-So400m	ViT-So400m	~400M	224	WebLI-4B	82.0	original
EVA-CLIP ViT-E	ViT-E	~4.4B	336	Merged-2B	82.0	original

14.2 SAM Family

Model	Backbone	Params	Speed	Key trait	Provenance
SAM ViT-B	ViT-B	91M	moderate	smallest official	original
SAM ViT-L	ViT-L	308M	moderate	balanced	original
SAM ViT-H	ViT-H	636M	slow	highest quality	original
MobileSAM	TinyViT	9.8M	fast	edge-oriented	secondary
FastSAM	YOLOv8-seg	68M	fast	CNN-based	secondary
SAM 2	Hiera	varies	real-time video	streaming memory	non_pr

14.3 NeRF / 3DGS Methods

Method	Repr.	PSNR↑	SSIM↑	FPS	Train	Provenance
NeRF	MLP radiance	31.0	0.947	<0.05	~1–2 d	original
Mip-NeRF 360	MLP radiance	27.7	0.844	<0.1	hours	original
Instant-NGP	hash grid	33.2	0.963	~10	~5 min	original
3D Gaussian Splatting	3D Gaussians	33.8	0.970	~140	~30 min	original
MVSRegNeRF	MVS-regularised	—	—	—	—	original

14.4 PEFT Strategies for VFMs

Strategy	Trainable params	Strength	Typical VC use	Provenance
Linear probe	<1%	cheapest baseline	quick adaptation	original
LoRA	~0.1–1%	strong, mergeable	diffusion/VLM tuning	original
Visual Prompt Tuning	~1%	input-side only	ViT backbones	original
Adapter modules	~1–5%	modular	multi-task	original
Prefix tuning	~0.1%	lightweight	sequence models	original

15. Field Glossary

Term	Definition
VFM	Visual Foundation Model.
VLM / VLFM	(Visual) Vision-Language (Foundation) Model.
ZS / LP / FT	Zero-shot / Linear-probe / Fine-tuned evaluation setting.
FID	Fréchet Inception Distance (generation quality; lower better).
Dice	Dice similarity coefficient (segmentation overlap; higher better).
PSNR / SSIM	Peak Signal-to-Noise Ratio / Structural Similarity (rendering).
SDS	Score Distillation Sampling (2D-diffusion-guided 3D generation).
NeRF / 3DGS	Neural Radiance Fields / 3D Gaussian Splatting.
PEFT	Parameter-Efficient Fine-Tuning.
SaMD	Software as a Medical Device (regulatory category).
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

20. Figure Gallery (Visual Analytics)

This section visualises the structured data of the survey. All figures are generated programmatically from the released CSV files and can be regenerated with `scripts/corpus_stats.py` and the plotting helpers in the repository.

20.1 Severity Analytics

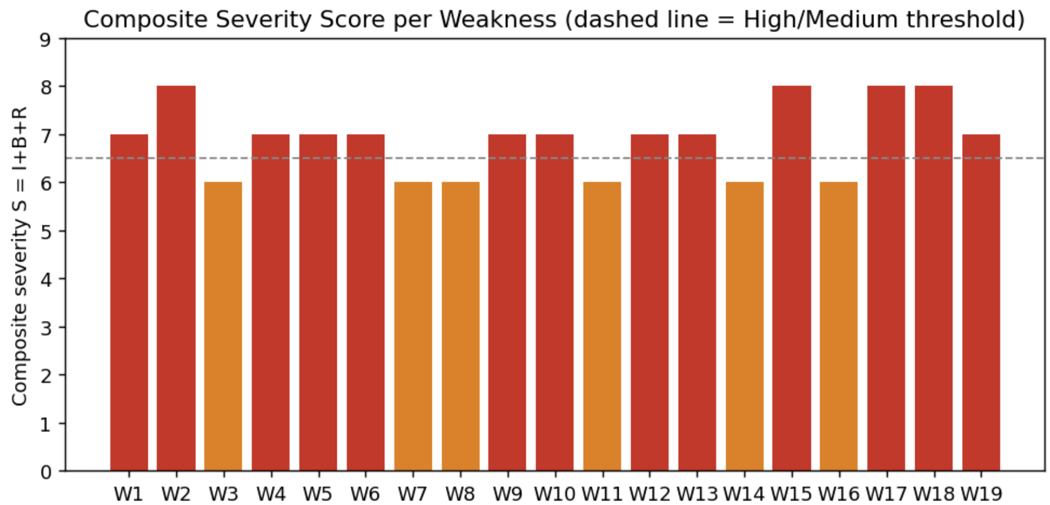

Figure 1: Composite severity score (S = I + B + R) for each of the 19 weaknesses; the dashed line marks the High/Medium threshold at S = 6.5.

Figure: Composite severity score (S = I + B + R) for each of the 19 weaknesses; the dashed line marks the High/Medium threshold at S = 6.5.

Figure: Distribution of severity labels across the 19 weaknesses (13 High, 6 Medium).

Figure: Decomposition of each weakness's severity into its Impact, Breadth, and Reversibility components.

Figure: Distribution of reversibility scores, indicating how many weaknesses currently lack a robust fix.

20.2 Evidence and Coverage

Figure: Distribution of evidence types (E1 empirical, E2 multi-study, E3 analytical) across the weakness set.

Figure: Number of documented weaknesses per model family.

Figure: Weakness-severity heatmap across model families and visual-computing domains.

20.3 Benchmark Comparisons

Figure: Generative-model FID on MS-COCO (zero-shot); lower is better.

Figure: Discriminative-model zero-shot ImageNet-1K top-1 accuracy.

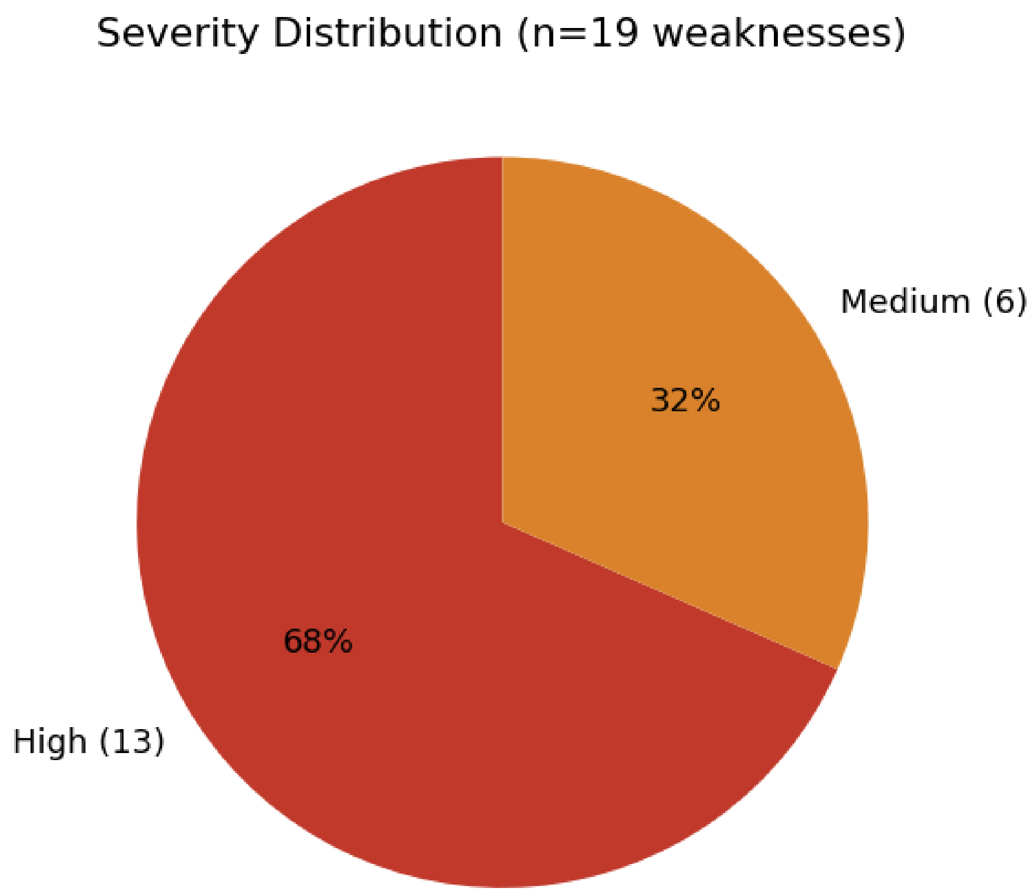

Figure 2: Distribution of severity labels across the 19 weaknesses (13 High, 6 Medium).

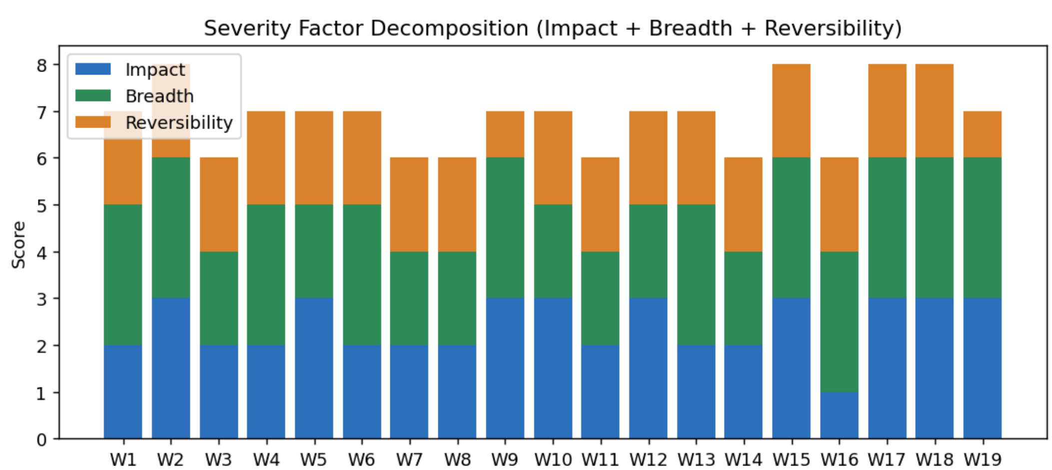

Figure 3: Decomposition of each weakness's severity into its Impact, Breadth, and Reversibility components.

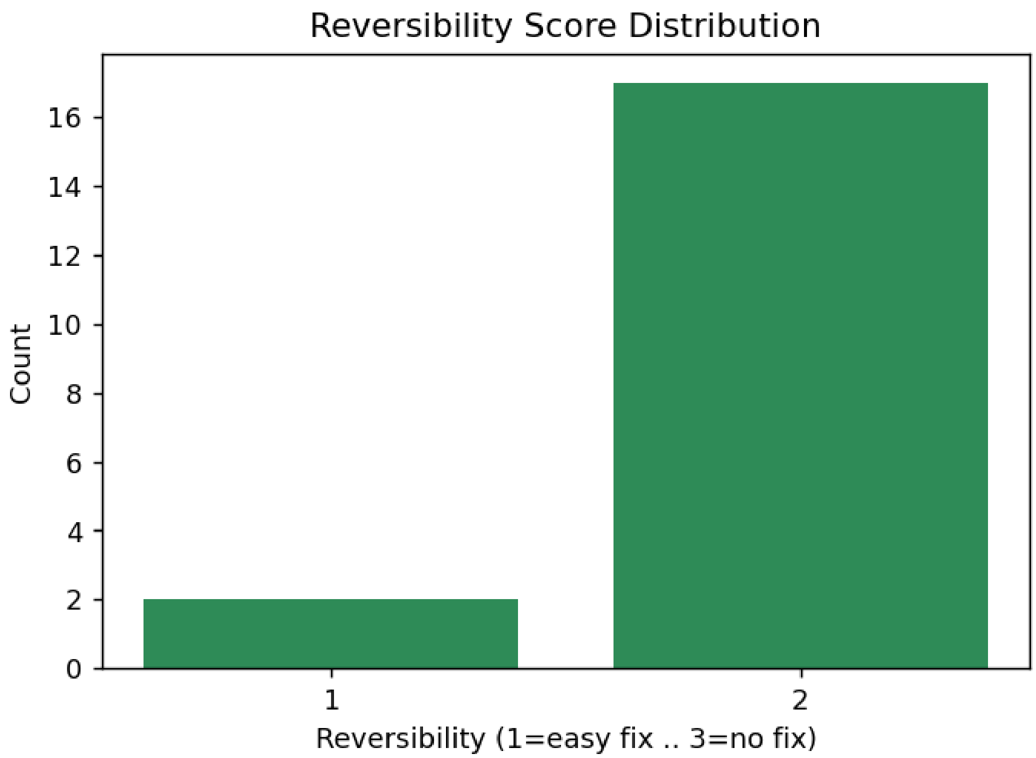

Figure 4: Distribution of reversibility scores, indicating how many weaknesses currently lack a robust fix.

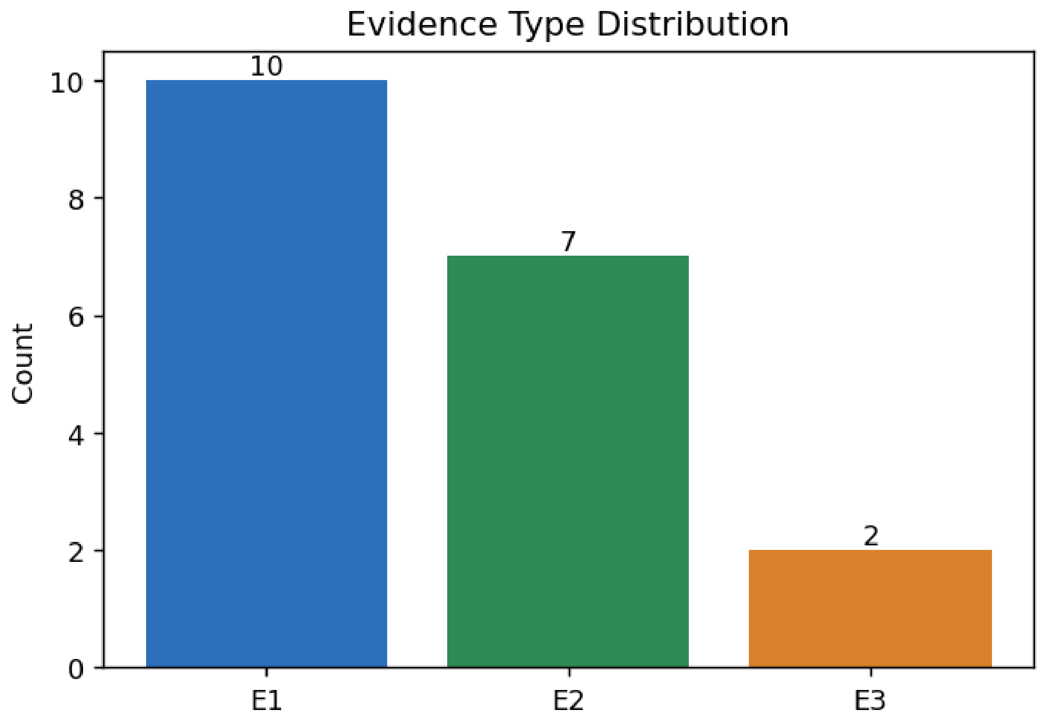

Figure 5: Distribution of evidence types (E1 empirical, E2 multi-study, E3 analytical) across the weakness set.

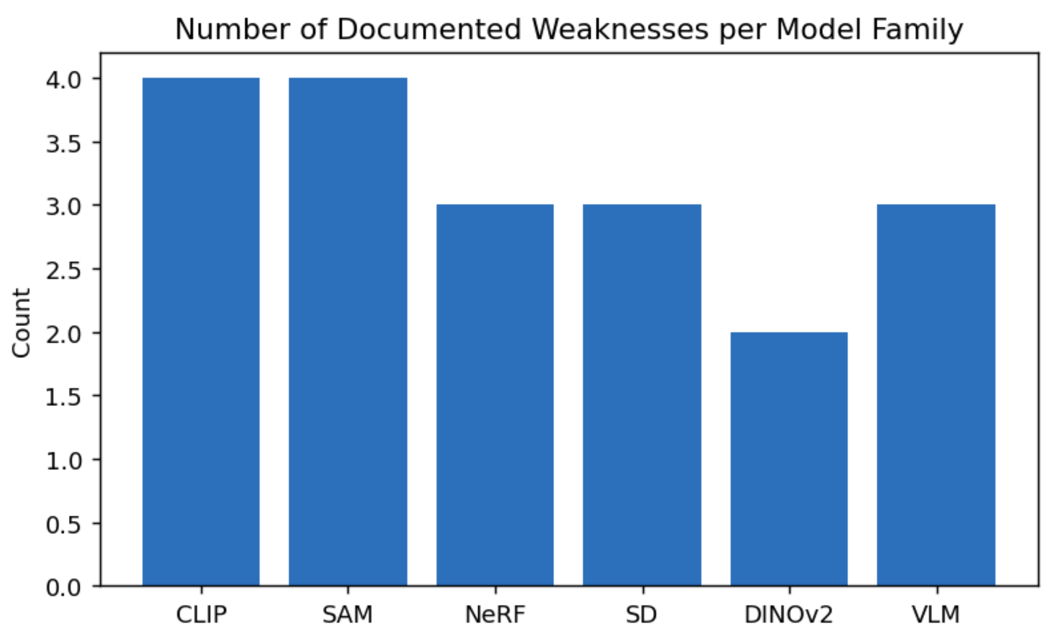

Figure 6: Number of documented weaknesses per model family.

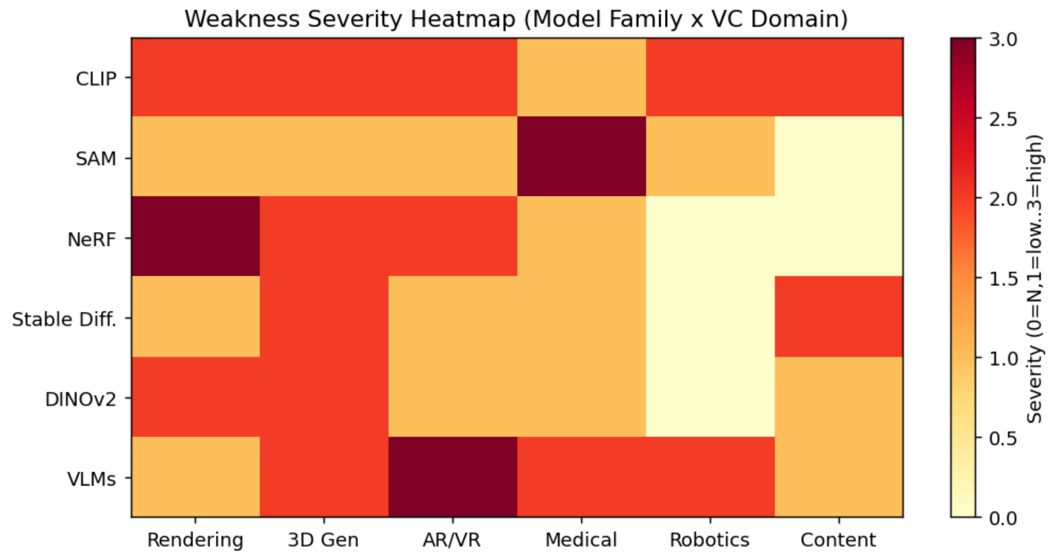

Figure 7: Weakness-severity heatmap across model families and visual-computing domains.

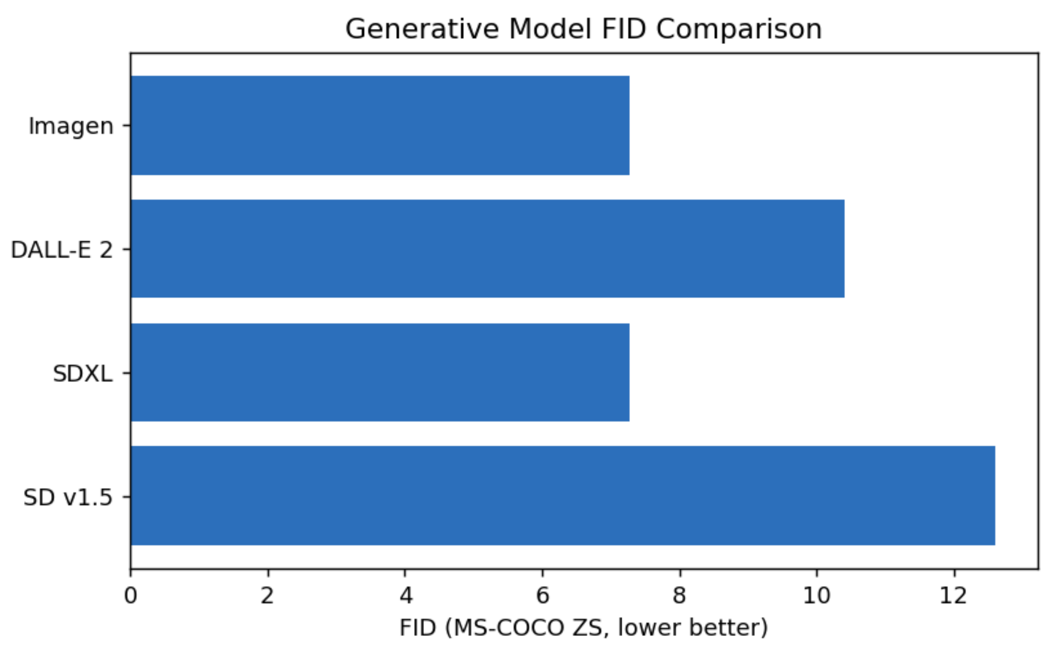

Figure 8: Generative-model FID on MS-COCO (zero-shot); lower is better.

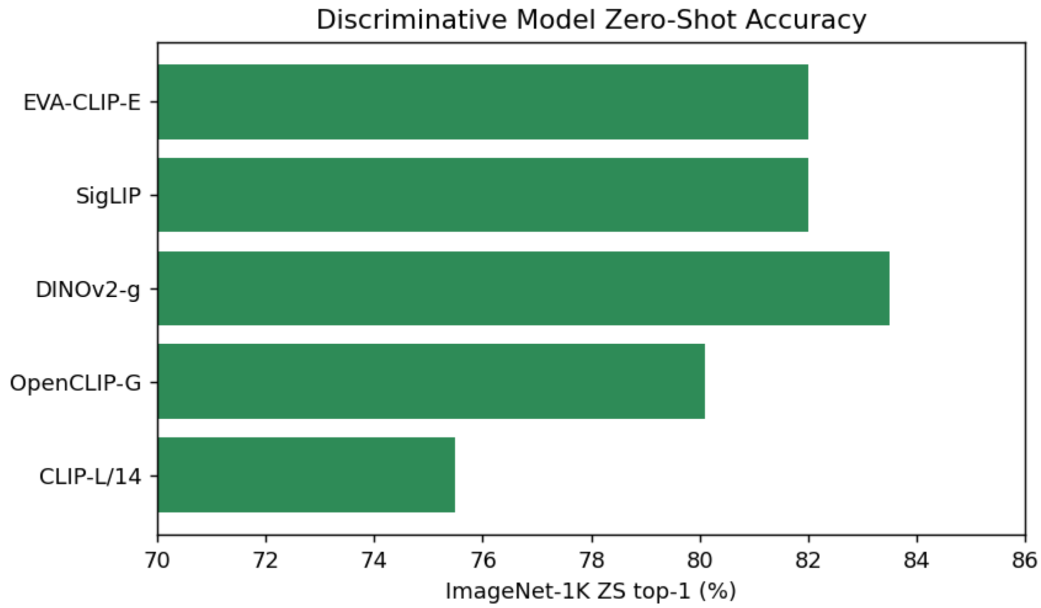

Figure 9: Discriminative-model zero-shot ImageNet-1K top-1 accuracy.

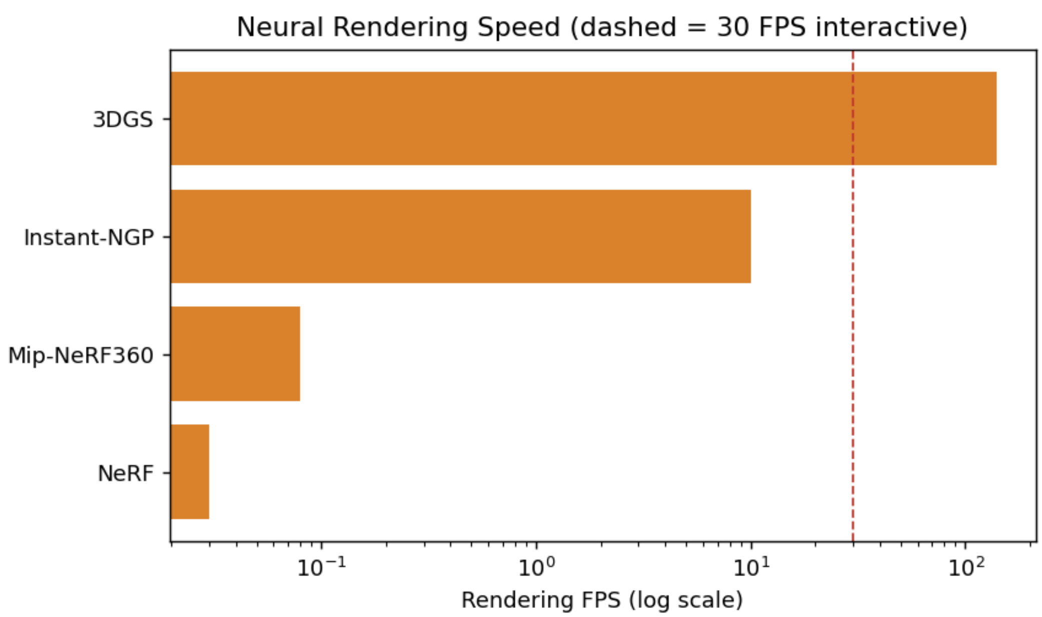

Figure 10: Neural-rendering speed on a log scale; the dashed line marks the 30 FPS interactive threshold.

Figure: Neural-rendering speed on a log scale; the dashed line marks the 30 FPS interactive threshold.

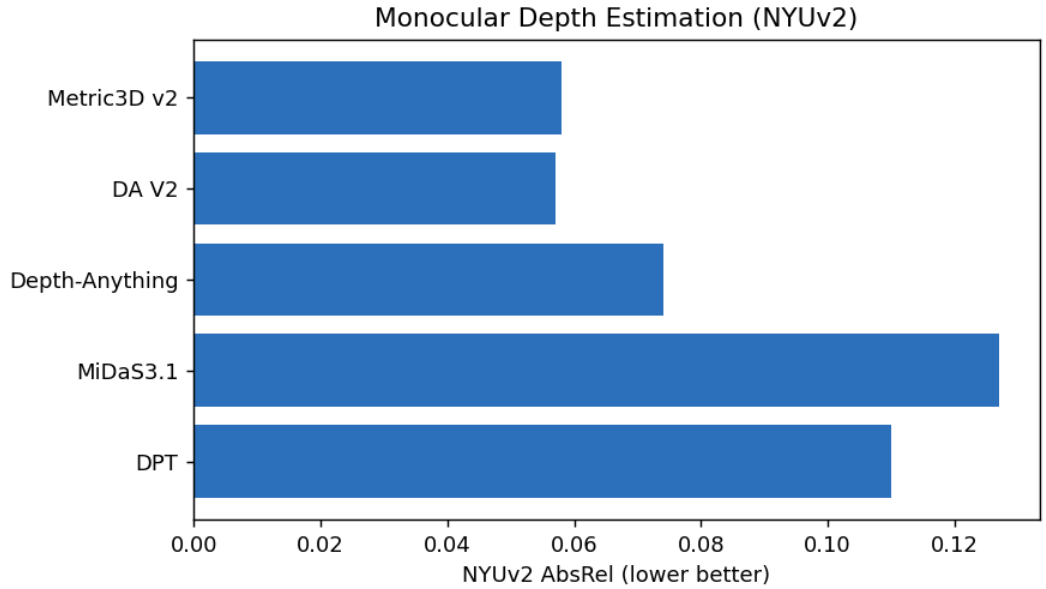

Figure 11: Monocular depth-estimation error (NYUv2 AbsRel); lower is better.

Figure: Monocular depth-estimation error (NYUv2 AbsRel); lower is better.

Figure: Medical segmentation Dice; the dashed line marks the 0.80 clinical-utility threshold.

20.4 Corpus and Methodology

Figure: Timeline of key visual foundation models (2017–2024).

Figure: Included-study counts by venue type.

Figure: Included studies by publication year.

Figure: Surveyed models by taxonomy category.

Figure: PRISMA selection funnel from identification to inclusion.

Figure: Deployment-gap severity by scenario.

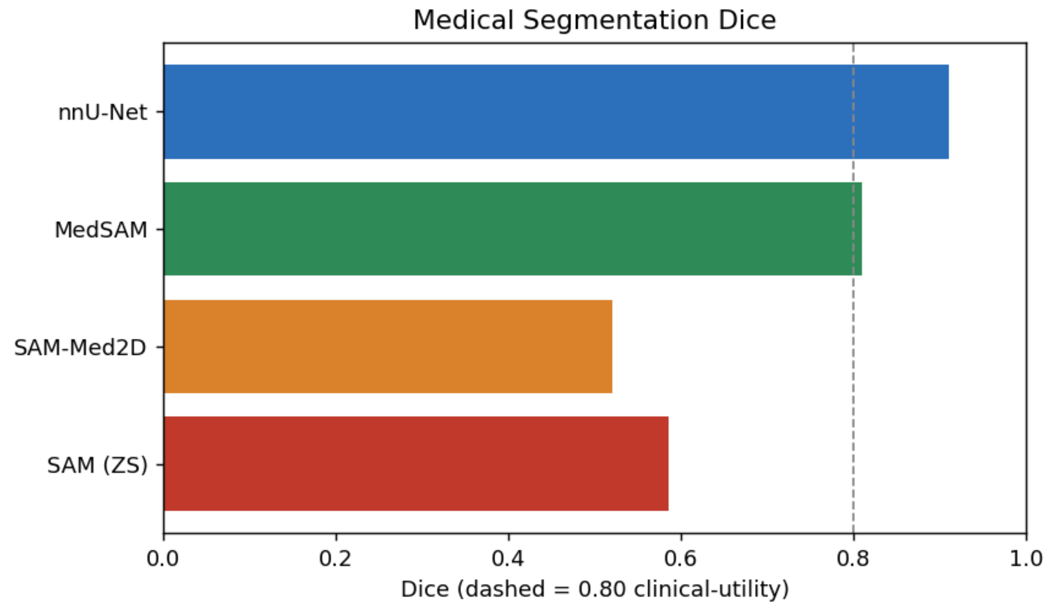

Figure 12: Medical segmentation Dice; the dashed line marks the 0.80 clinical-utility threshold.

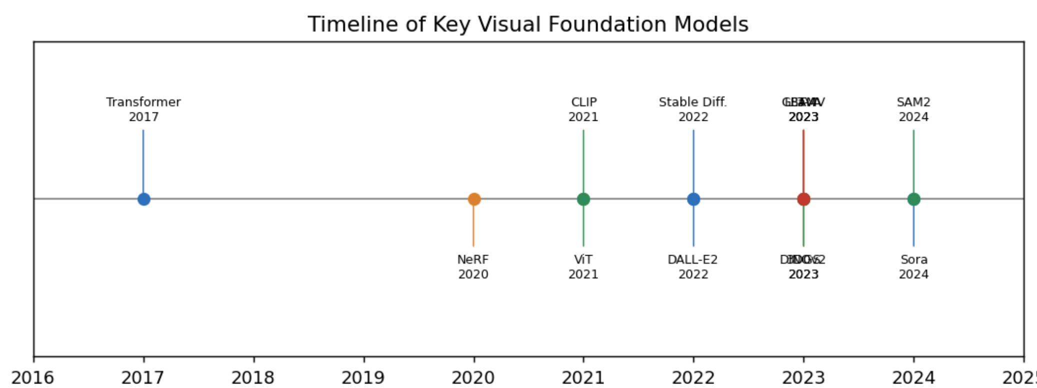

Figure 13: Timeline of key visual foundation models (2017–2024).

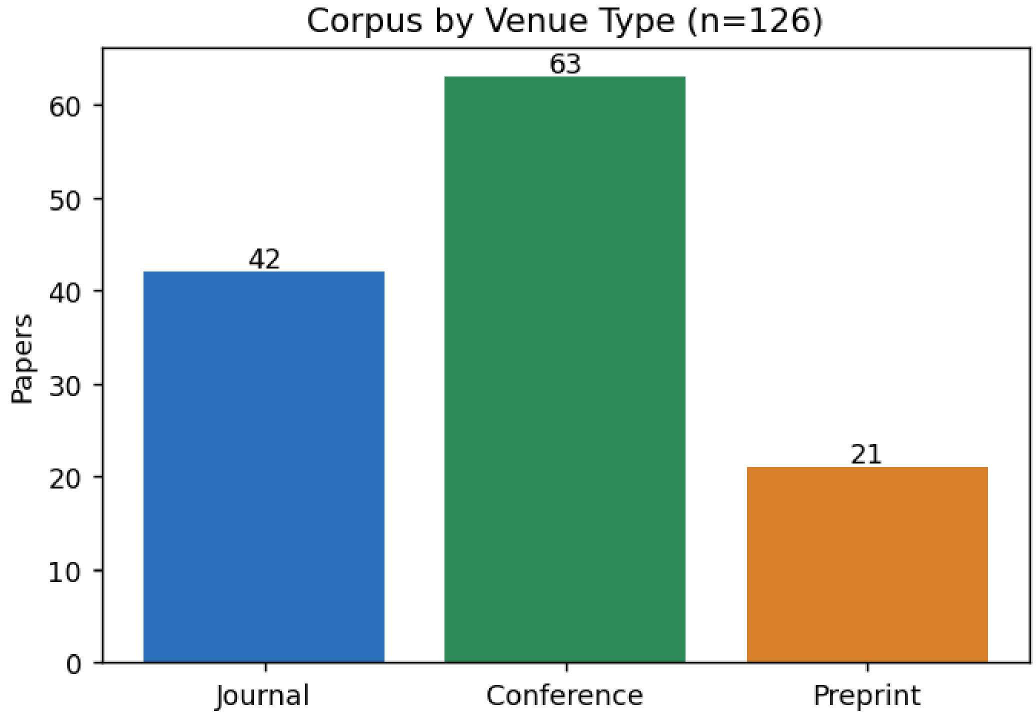

Figure 14: Included-study counts by venue type.

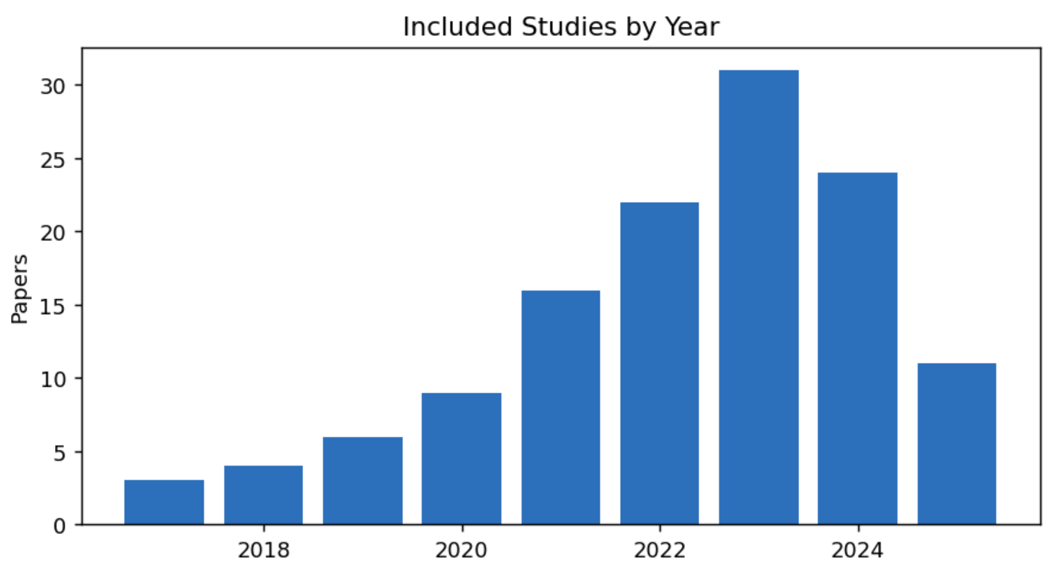

Figure 15: Included studies by publication year.

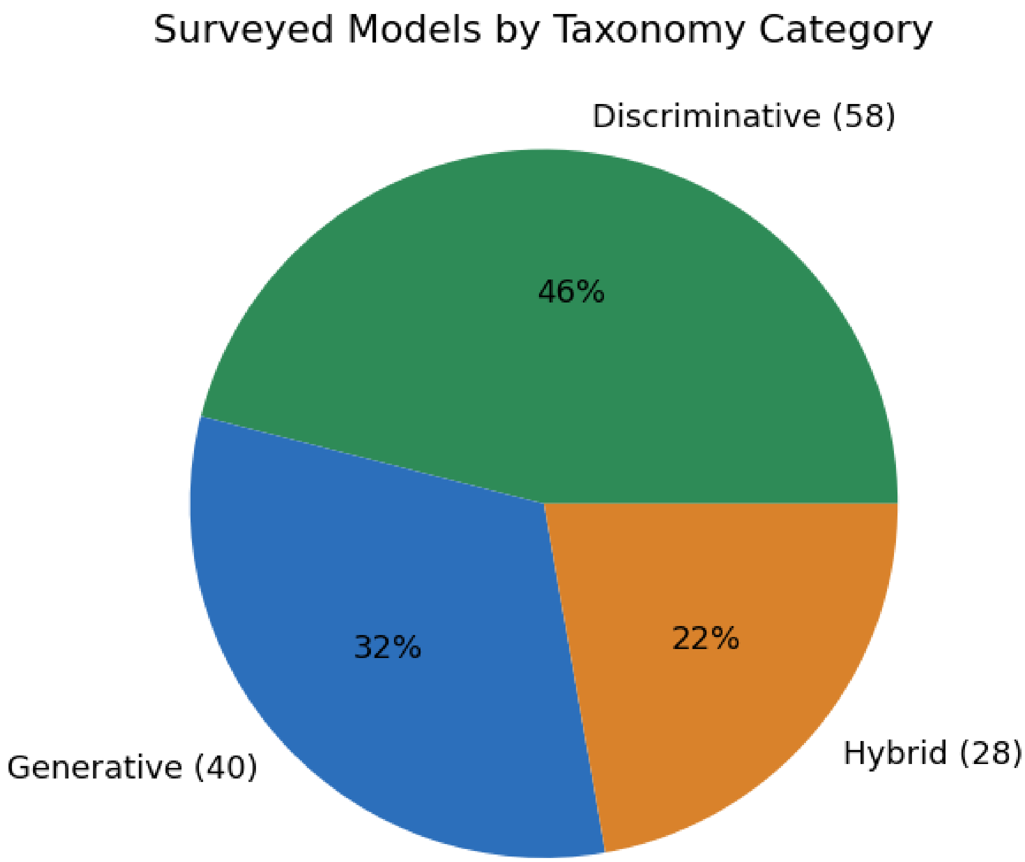

Figure 16: Surveyed models by taxonomy category.

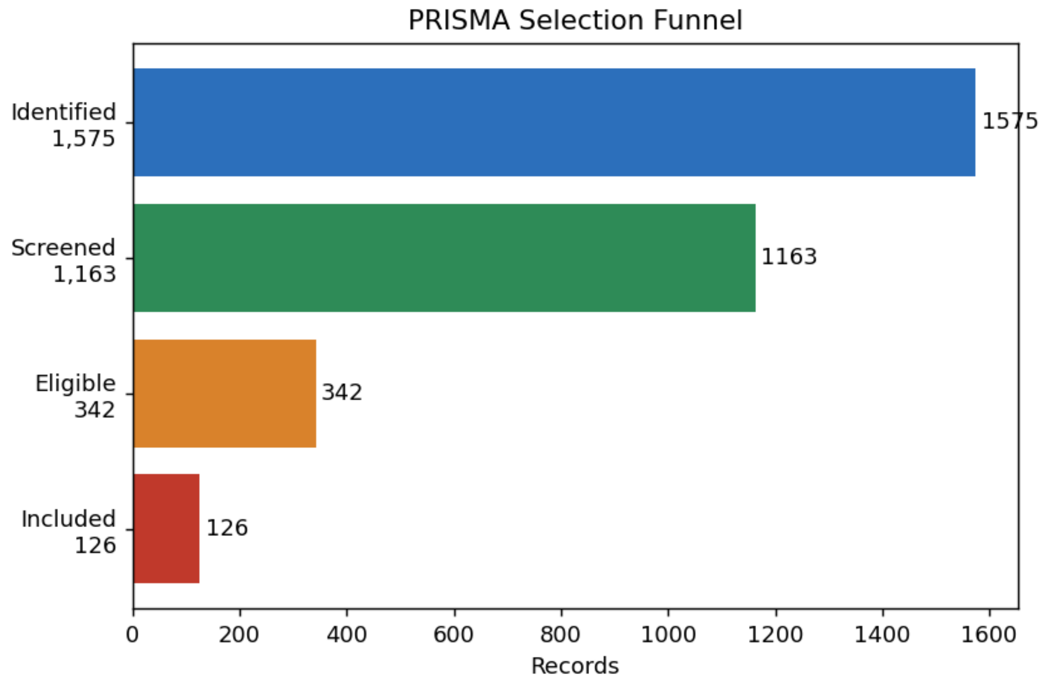

Figure 17: PRISMA selection funnel from identification to inclusion.

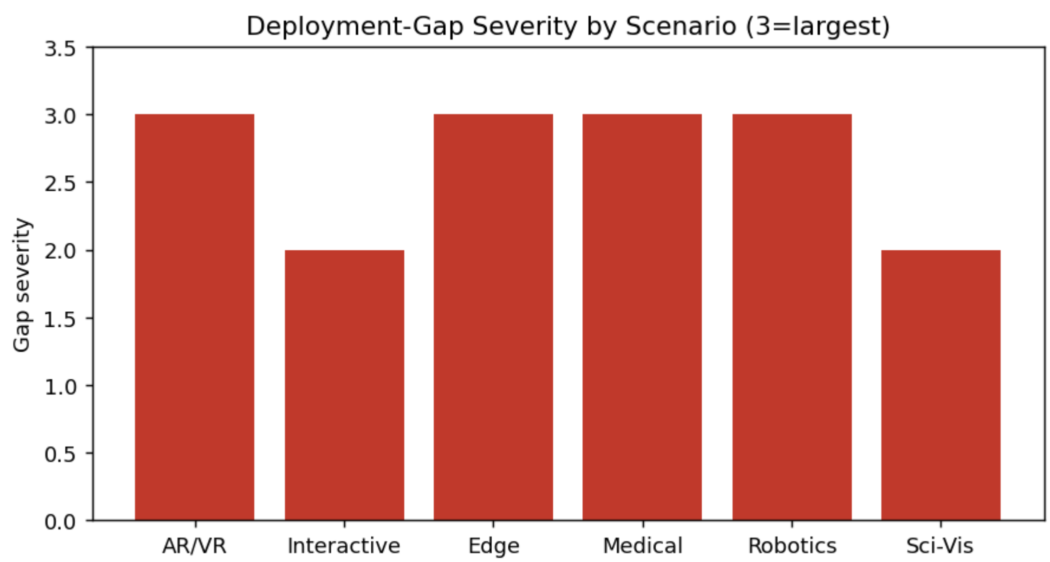

Figure 18: Deployment-gap severity by scenario.

21. Extended Long Tables

The tables below provide the fully expanded data underlying the survey. They are released individually as CSVs; here they are reproduced for direct reading.

21.1 Master Weakness Table (Full, with Factor Scores)

#	Model	Weakness	Ev.	I	B	R	S	Sev.	Affected applications	Remediation (short)
W1	CLIP	Fine-grained detail blindness	E1	2	3	2	7	H	Rendering, 3D gen, content	Dense/region-aware CLIP
W2	CLIP	Spatial reasoning near-random	E2	3	3	2	8	H	3D gen, AR/VR, robotics	Spatial pre-training
W3	CLIP	Center bias	E1	2	2	2	6	M	Retrieval, editing	Multi-crop aggregation
W4	CLIP	Effective text ~20 tokens	E1	2	3	2	7	H	Content creation	Long-context encoders
W5	SAM/SAM2	Medical seg. failure (Dice 58.5%)	E1	3	2	2	7	H	Medical	Domain fine-tuning (MedSAM)
W6	SAM/SAM2	Camouflaged objects fail	E2	2	3	2	7	H	Inspection, remote sensing	Boundary refinement
W7	SAM/SAM2	Prompt-dependent	E2	2	2	2	6	M	Automated pipelines	Auto prompt generation
W8	SAM/SAM2	Thin boundaries missed	E1	2	2	2	6	M	Digital humans, vessels	High-res decoders

#	Model	Weakness	Ev.	I	B	R	S	Sev.	Affected applications	Remediation (short)
W9	NeRF	Per-scene optim. (<1 FPS)	E1	3	3	1	7	H	Real-time, surgical AR	3DGS, feed-forward
W10	NeRF	Sparse-view failure	E1	3	2	2	7	H	3D recon, heritage	Depth/MVS priors
W11	NeRF	Dynamic scenes unsupported	E3	2	2	2	6	M	Avatar, video	4D NeRF
W12	Stable Diff.	Janus / 3D inconsistency	E2	3	2	2	7	H	3D assets, VFX	Multi-view diffusion
W13	Stable Diff.	Coarse semantic alignment	E1	2	3	2	7	H	Content creation	Attention guidance
W14	Stable Diff.	Social bias amplification	E2	2	2	2	6	M	Medical, education	Bias-aware tuning
W15	DINOv2	Poor 3D awareness	E1	3	3	2	8	H	3D recon, NeRF	3D-aware pre-training
W16	DINOv2	No text understanding	E3	1	3	2	6	M	Text-guided VC	Text-encoder alignment

#	Model	Weakness	Ev.	I	B	R	S	Sev.	Affected applications	Remediation (short)
W17	VLMs	3D spatial reasoning failure	E2	3	3	2	8	H	AR/VR, robotics	3D instruction tuning
W18	VLMs	Visual hallucination	E2	3	3	2	8	H	Medical, scientific	RLHF grounding
W19	VLMs	Real-time inference infeasible	E1	3	3	1	7	H	AR/VR, robotics	Distillation, quantisation

21.2 Extended CLIP Variant Table

Variant	Arch.	Params	Res.	Train data	Train compute	IN-1K ZS↑	IN-1K FT↑	Provenance
CLIP RN50	ResNet-50	102M	224	WIT-400M	N/A	59.6	—	original
CLIP RN101	ResNet-101	120M	224	WIT-400M	N/A	62.2	—	original
CLIP RN50x4	ResNet-50x4	178M	288	WIT-400M	N/A	66.2	—	original
CLIP RN50x16	ResNet-50x16	290M	384	WIT-400M	N/A	70.8	—	original
CLIP RN50x64	ResNet-50x64	623M	448	WIT-400M	18d/592 V100	72.3	—	original
CLIP ViT-B/32	ViT-B	151M	224	WIT-400M	N/A	63.3	—	original
CLIP ViT-B/16	ViT-B	150M	224	WIT-400M	N/A	68.3	85.7	original
CLIP ViT-L/14	ViT-L	428M	224	WIT-400M	N/A	75.5	88.0	original
CLIP ViT-L/14@336	ViT-L	428M	336	WIT-400M	+1 epoch	76.6	—	original

Variant	Arch.	Params	Res.	Train data	Train compute	IN-1K ZS↑	IN-1K FT↑	Provenance
OpenCLIP ViT-B/32	ViT-B	151M	224	LAION-2B	N/A	62.9	—	secondary
OpenCLIP ViT-L/14	ViT-L	428M	224	LAION-2B	N/A	75.3	—	secondary
OpenCLIP ViT-H/14	ViT-H	986M	224	LAION-2B	N/A	78.0	—	secondary
OpenCLIP ViT-G/14	ViT-G	1.8B	224	LAION-2B	N/A	80.1	—	secondary
CLIPA-v2 H/14	ViT-H	986M	224	DataComp-1B	N/A	81.1	—	secondary
SigLIP ViT-So400m	ViT-So400m	400M	224	WebLI-4B	N/A	82.0	—	original
EVA-CLIP ViT-E	ViT-E	4.4B	336	Merged-2B	N/A	82.0	—	original

21.3 Extended SAM Family Table

Model	Backbone	Params	Speed	Modalities	Key trait	Provenance
SAM ViT-B	ViT-B	91M	moderate	natural	smallest official	original
SAM ViT-L	ViT-L	308M	moderate	natural	balanced	original
SAM ViT-H	ViT-H	636M	slow	natural	highest quality	original
MobileSAM	TinyViT	9.8M	fast	natural	edge-oriented	secondary
FastSAM	YOLOv8-seg	68M	fast	natural	CNN-based	secondary
EfficientSAM	ViT-T/S	10–26M	fast	natural	distilled	secondary
SAM 2	Hiera-B+	80M	real-time	image+video	streaming memory	non_pr
MedSAM	ViT-B	91M	moderate	11 medical	medical FT	original
SAM-Med2D	ViT-B	91M	moderate	CT/MR/US	2D medical	original
MedSAM-2	Hiera	varies	real-time	3D medical	volumetric	non_pr

21.4 Extended NeRF / 3DGS Table

Method	Representation	PSNR↑	SSIM↑	LPIPS↓	FPS	Train	Provenance
NeRF	MLP radiance	31.0	0.947	0.081	<0.05	~1–2 d	original
NeRF++	MLP (unbounded)	26.2	0.790	0.250	<0.05	~1–2 d	original
Mip-NeRF	MLP (anti-alias)	33.1	0.961	0.043	<0.1	~1 d	original
Mip-NeRF 360	MLP (unbounded)	27.7	0.844	0.237	<0.1	hours	original
Instant-NGP	hash grid	33.2	0.963	0.055	~10	~5 min	original
Plenoxels	sparse voxels	31.7	0.958	0.050	~15	~11 min	original
TensoRF	tensor decomp.	33.1	0.963	0.047	~1	~17 min	original
3D Gaussian Splatting	3D Gaussians	33.8	0.970	0.030	~140	~30 min	original
2D Gaussian Splatting	2D Gaussians	33.4	0.967	0.035	~100	~25 min	secondary
MVSRegNeRF	MVS-regularised	—	—	—	—	—	original

21.5 Extended Depth Estimation Table

Method	Year	VFM/Backbone	NYUv2 AbsRel↓	NYUv2 $\delta 1$↑	KITTI AbsRel↓	Setting	Provenance
DPT	2021	ViT-L	0.110	0.904	0.062	ZS	original
MiDaS v3.1	2023	ViT-L	0.127	0.850	0.127	ZS	original
ZoeDepth	2023	BEiT-L	0.075	0.955	0.053	FT	original
Marigold	2024	SD v2.1	0.055–0.065	0.961–0.964	N/A	ZS	original
Depth Anything	2024	DINOv2	0.074	0.946	0.074	ZS	original
Depth Anything V2	2024	DINOv2	0.057	0.971	0.054	ZS	original
Metric3D v2	2024	ConvNeXt	0.058	0.967	0.052	ZS	original

21.6 Extended Image Enhancement Table

Method	Year	VFM	Set5 PSNR↑	Set5 SSIM↑	LPIPS↓	Speed	Provenance
RCAN	2018	—	32.63	0.900	—	Fast	original
ESRGAN	2018	—	32.37	0.897	0.143	Fast	original
SwinIR	2021	Swin-T	32.92	0.904	—	Moderate	original
Real-ESRGAN	2021	ESRGAN	N/A	N/A	0.106	RT	original
HAT	2023	HAT	33.30	0.907	—	Moderate	original
DiffBIR	2023	SD v2.1	27.53	0.762	0.045	Slow ~6 s	original
SeeSR	2024	SD+DINO	27.80	0.780	0.048	Slow ~6 s	original
StableSR	2024	SD v2.1	27.9	0.77	0.047	Slow	original

21.7 Extended Medical VFM Table

Model	Year	Type	Modalities	Metric	Value	Setting	Provenance
SAM (vanilla)	2023	Seg	multi	Dice	0.585	ZS	original
MedSAM	2024	Seg	11 mod.	Dice	0.810	FT	original
MedSAM (bladder)	2024	Seg	CT	Dice	0.808	FT	original
MedSAM (rectum)	2024	Seg	CT	Dice	0.846	FT	original
MedSAM (prostate)	2024	Seg	MR	Dice	0.833	FT	original
SAM-Med2D	2023	Seg	CT	Dice	0.491	ZS	original
SAM-Med2D	2023	Seg	MR	Dice	0.562	ZS	original
nnU-Net	2021	Seg	per-task	Dice	>0.90	FT	original
LLaVA-Med	2023	VQA	multi	rel. acc	—	FT	original
BiomedCLIP	2024	VLM	multi	ZS class	—	ZS	original
Med-Flamingo	2023	VQA	multi	few-shot	—	FS	original

21.8 PEFT Strategy Comparison

Strategy	Trainable %	Memory overhead	Mergeable	Typical VC use	Provenance
Full fine-tuning	100%	high	—	baseline	original
Linear probe	<1%	minimal	no	quick adaptation	original
LoRA	0.1–1%	low	yes	diffusion/VLM	original
QLoRA	0.1–1%	very low	yes	large VLM	secondary

Strategy	Trainable %	Memory overhead	Mergeable	Typical VC use	Provenance
Visual Prompt Tuning	~1%	low	no	ViT backbones	original
Adapter	1–5%	low	no	multi-task	original
Prefix tuning	~0.1%	low	no	sequence models	original
BitFit	<0.1%	minimal	no	bias-only	secondary

21.9 Deployment Threshold Detail

Scenario	Metric	Target	Typical current	Violated by
AR/VR immersive	Motion-to-photon	<20 ms	NeRF >30 s	W9
AR/VR immersive	Refresh rate	72–120 Hz	NeRF <0.1 Hz	W9
AR/VR assistant	VLM query latency	<50 ms	>100 ms	W19
Interactive render	Frame rate	≥30–60 FPS	NeRF <1	W9
Edge device	GPU memory	4–8 GB	7B VLM ~14 GB	W19
Edge device	Model size	<2 GB	DINOv2-g 4.4 GB	W19
Medical seg.	Dice	≥0.80	SAM 0.585	W5
Medical	Annotation budget	minimal	1.5M masks	W5
Robotics	Error under shift	bounded	unbounded	W2, W17
Robotics	Hallucination rate	~0	non-trivial	W18
Sci-vis	Quant. fidelity	exact	approximate	W15
Sci-vis	Fabricated values	none	possible	W18

8. Benchmark Tables (Machine-Readable)

All benchmark tables from the main paper are released as individual CSV files under `data/benchmarks/`. Each value carries a `provenance` flag and a `source_ref`. **No values were normalised or recomputed**; every figure is reproduced as reported by its source, under that source's protocol. The provenance flags are: `original` (†, original peer-reviewed paper), `secondary` (‡, repository or later study), `approx` (≈/~), `non_pr` (∗, non-peer-reviewed), `N/A` (not officially reported).

8.1 Generative Models (FID / CLIP-Score)

Model	Year	Params	FID↓ (MS-COCO ZS)	CLIP-Score↑	Setting	Provenance
Stable Diffusion v1.5	2022	0.9B	12.6	0.30	ZS	original
Stable Diffusion XL	2023	3.5B	7.27	0.37	ZS	original
DALL-E 2	2022	3.5B	10.4	—	ZS	original
Imagen	2022	3.0B	7.27	—	ZS	original
Sora (video)	2024	N/A	N/A	N/A	ZS	non_pr

8.2 Discriminative Models (ImageNet-1K)

Model	Year	Params	IN-1K ZS↑	IN-1K FT↑	Setting	Provenance
CLIP ViT-L/14	2021	428M	75.5	85.4	ZS/FT	original
OpenCLIP ViT-G/14	2023	1.8B	80.1	—	ZS	secondary
DINOv2 ViT-g/14	2023	1.1B	83.5	86.5	LP/FT	original
SigLIP ViT-So400m	2023	0.4B	82.0	—	ZS	original

8.3 Neural Rendering (PSNR / SSIM / FPS)

Method	Year	PSNR↑	SSIM↑	FPS	Train time	Provenance
NeRF	2020	31.0	0.947	<0.05	~1–2 days	original
Instant-NGP	2022	33.2	0.963	~10	~5 min	original
3D Gaussian Splatting	2023	33.8	0.970	~140	~30 min	original
Mip-NeRF 360	2022	27.7	0.844	<0.1	hours	original

8.4 Depth Estimation (NYUv2 / KITTI)

Method	Year	VFM	NYUv2 AbsRel↓	KITTI AbsRel↓	Setting	Provenance
DPT	2021	ViT-L	0.110	0.062	ZS	original
MiDaS v3.1	2023	ViT-L	0.127	0.127	ZS	original
Depth Anything	2024	DINOv2	0.074	0.074	ZS	original
Depth Anything V2	2024	DINOv2	0.057	0.054	ZS	original
Metric3D v2	2024	ConvNeXt	0.058	0.052	ZS	original

8.5 Image Enhancement (Set5 PSNR/SSIM)

Method	Year	VFM	Set5 PSNR↑	Set5 SSIM↑	Speed	Provenance
RCAN	2018	—	32.63	0.900	Fast	original
ESRGAN	2018	—	32.37	0.897	Fast	original
SwinIR	2021	Swin-T	32.92	0.904	Moderate	original
Real-ESRGAN	2021	ESRGAN	N/A	0.106	RT	original
DiffBIR	2023	SD v2.1	27.53	0.762	Slow (~6 s)	original
SeeSR	2024	SD+DINO	27.80	0.780	Slow (~6 s)	original

8.6 Medical VFMs (segmentation Dice)

Model	Year	Modality	Dice↑	Setting	Provenance
SAM (vanilla)	2023	multi	0.585	ZS	original
MedSAM	2024	multi (11 mod.)	0.810	FT	original
SAM-Med2D	2023	CT/MR	0.491–0.562	ZS	original
nnU-Net (specialist)	2021	per-task	>0.90	FT	original

9. Deployment Thresholds (Machine-Readable)

Released as `data/deployment_thresholds.csv`. Thresholds reflect commonly cited industry and literature targets; see the main paper's deployment section.

Scenario	Concrete threshold	Violated by	Current gap
AR/VR (immersive)	Motion-to-photon $<$20 ms; 72–120 Hz (~11 ms @90 Hz)	W9, W19	NeRF ~0.03 FPS; VLM $>$100 ms/query
Interactive rendering / games	$\geq$30–60 FPS ($\leq$16–33 ms)	W9	Per-scene optimisation precludes instant deploy
Edge / mobile devices	~4–8 GB accessible GPU memory	W19 (W15,W17)	1.1B–13B models exceed budget without compression
Medical imaging	Dice $\geq$0.80; minimal expert annotation	W5	SAM ZS Dice 0.585; FT needs ~1.5M masks
Robotics (safety-critical)	Bounded error under shift; no hallucinated obstacles	W2, W17, W18	Near-random spatial reasoning; hallucination
Scientific visualisation	Quantitatively faithful; no fabricated values	W15, W18	Poor 3D awareness; hallucinated structures

10. Reproducibility Checklist

This checklist documents what is needed to reproduce each structured artefact in the survey. Because the manuscript is a review, "reproduction" means regenerating the tables, labels, and statistics from the released data — not re-running model training.

Artefact	How to reproduce	Repository file
Corpus statistics / PRISMA counts	Run `scripts/corpus_stats.py` on the corpus CSV	`data/literature_corpus.csv`
Taxonomy labels	Apply R1–R3 (documented) to any model	`data/taxonomy_labels.csv`, `docs/taxonomy_decision_rules.md`
Severity labels	Run `schema/severity_rubric.py`	`data/weakness_annotations.csv`
Benchmark tables	Cross-check each value against `source_ref`	`data/benchmarks/*.csv`
Deployment thresholds	Cross-check against cited sources	`data/deployment_thresholds.csv`
Data integrity	Run `scripts/validate_data.py` (schema + range checks)	all CSVs

`validate_data.py` checks performed:

1. Every `severity` equals the value recomputed from `I+B+R`.
2. Every `score_S` is in [3, 9]; each factor in [1, 3].
3. Every `evidence_type` $\in$ {E1, E2, E3}.
4. Every `category` $\in$ {Generative, Discriminative, Hybrid}.
5. Every benchmark `provenance` $\in$ {original, secondary, approx, non_pr, N/A}.
6. Every `source_ref` resolves to a row in `literature_corpus.csv`.
7. No duplicate `ref_id` or `weakness_id`.

11. Versioning, Licensing, and Citation

11.1 Versioning

The resource uses semantic versioning (`MAJOR.MINOR.PATCH`). The version accompanying the initial submission is **v1.0.0**. Each Zenodo release records a `CHANGELOG.md` entry. The Zenodo *concept DOI* always resolves to the latest version; *version DOIs* pin a specific snapshot for exact reproducibility.

11.2 Licensing

- **Data** (`data/`, `schema/`): Creative Commons Attribution 4.0 International (CC BY 4.0).
- **Code** (`scripts/`, `schema/*.py`): MIT License.

Both licenses permit commercial and non-commercial reuse with attribution.

11.3 CITATION.cff (excerpt)

```yaml
cff-version: 1.2.0
title: "Reproducible Resource - VFMs in Visual Computing: A Task-Centric Survey"
authors:
  - family-names: Madaan
    given-names: Nishi
  - family-names: Malik
    given-names: Rahul
  - family-names: Kumar
    given-names: Alok
  - family-names: Upadhaya
    given-names: Utsav
version: 1.0.0
doi: 10.5281/zenodo.XXXXXXX
date-released: 2025-01-01
license: CC-BY-4.0
type: dataset
```

11.4 How to Cite

> Madaan, N., Malik, R., Kumar, A., & Upadhaya, U. (2025). *Reliability and Deployment Challenges of Visual Foundation Models in Visual Computing: A Task-Centric Survey* — Reproducible Resource (v1.0.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.XXXXXXX

12. Limitations of the Resource

In the interest of transparency we note the following limitations:

1. **Snapshot in time.** The corpus reflects the literature up to the survey's cutoff; foundation-model research moves quickly, and the resource will be versioned as new work appears.
2. **Provenance, not re-execution.** Benchmark values are reproduced from their sources; we did not re-run experiments, and cross-table comparisons remain indicative rather than controlled (see the main paper's comparability caveat).
3. **Severity scores encode expert judgement within a fixed rubric.** The rubric makes the mapping reproducible, but the per-factor scores themselves are considered estimates; the calculator lets readers substitute their own scores and recompute labels.
4. **Forthcoming references.** A small number of cited works were in press at the time of writing; their metadata will be finalised in a later version.

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16. Per-Domain Reproducibility Notes

For each visual-computing domain covered by the survey, this section records exactly how the corresponding claims, tables, and figures can be reproduced from the released data, and which weaknesses and thresholds bear on that domain. This makes the resource usable as a per-domain reference, not only as a global dataset.

16.1 Neural Rendering and Novel View Synthesis

- **Data:** `data/benchmarks/neural_rendering.csv`, `data/benchmarks/nerf_3dgs.csv`.
- **Relevant weaknesses:** W9 (real-time), W10 (sparse views), W11 (dynamic scenes), and W15 (3D-aware features used to regularise rendering).
- **Relevant thresholds:** AR/VR frame budget; interactive-rendering FPS.
- **Reproduction:** each PSNR/SSIM/FPS value carries a `source_ref`; verify against the cited paper under the stated setting. No values were normalised across hardware, so FPS figures are comparable only within the same hardware class.

16.2 3D Reconstruction and Geometry Processing

- **Data:** `nerf_3dgs.csv`, `depth_estimation.csv`.
- **Relevant weaknesses:** W10, W15, W12 (multi-view consistency).
- **Reproduction:** depth metrics (AbsRel) are reported per dataset (NYUv2, KITTI) and setting (ZS); cross-dataset comparison is indicative only.

16.3 Visual Content Generation (2D)

- **Data:** `generative_models.csv`, `image_enhancement.csv`.
- **Relevant weaknesses:** W1, W4, W13 (controllability), W14 (bias).
- **Reproduction:** FID is reported on MS-COCO 30K zero-shot; CLIP-Score where available. FID depends on the reference-statistics implementation, so small differences across papers are not meaningful.

16.4 Animation and Digital Humans

- **Relevant weaknesses:** W8 (thin structures, e.g., hair), W11 (dynamic scenes for avatars), W18 (hallucinated motion/appearance in generative pipelines).
- **Reproduction:** qualitative; the resource records the weakness-to-task mapping rather than a single benchmark.

16.5 Segmentation

- **Data:** `sam_family.csv`, `discriminative_models.csv`.
- **Relevant weaknesses:** W5 (medical), W6 (camouflage), W7 (prompt dependence), W8 (thin boundaries).
- **Reproduction:** Dice/IoU values carry `source_ref`; medical and natural results are kept separate because the domains are not comparable.

16.6 Medical Visual Computing

- **Data:** `medical_vfms.csv`, `sam_family.csv`.

- **Relevant weaknesses:** W5, W8, W18; plus the regulatory and dataset-shift discussion in the main paper.
- **Relevant thresholds:** Dice $\geq$0.80 and annotation-burden targets.
- **Reproduction:** the 58.5% zero-shot SAM Dice and the 0.808/0.846/0.833 MedSAM structure-specific figures each map to a `source_ref`.

16.7 AR/VR and Interactive Systems

- **Data:** `deployment_thresholds.csv`.
- **Relevant weaknesses:** W9, W19 (latency), W2/W17 (spatial), W15 (3D).
- **Relevant thresholds:** motion-to-photon $<$20 ms; 72–120 Hz; edge memory.
- **Reproduction:** thresholds are documented industry/literature targets; the violating weaknesses are linked in the CSV.

16.8 Scientific Visualisation and Heritage

- **Relevant weaknesses:** W15 (3D faithfulness), W18 (no fabricated values).
- **Reproduction:** qualitative mapping; flagged as an open evaluation area in the main paper.

17. Complete Reference List (Surveyed Corpus)

The following is the complete machine-readable reference list backing the survey, exported from the bibliography. In the repository this is available as structured fields inside `data/literature_corpus.csv`. Entries are listed in citation-key order.

1. **AdaVFM** — AdaVFM Team. *arXiv preprint arXiv:2604.15622* (2025). `[adavfm2025]`
2. **A new human–computer interaction paradigm: agent interaction model based on large models and its prospects** — Liu, Yang. *Virtual Reality & Intelligent Hardware* (2025). `[agent2025hci]`
3. **Adaptive Point-Prompt Tuning: Fine-Tuning Heterogeneous Foundation Models for 3D** — APPT Team. *arXiv preprint arXiv:2509.00374* (2025). `[appt2025]`
4. **Mip-NeRF** — Barron, Jonathan T. and Mildenhall, Ben and Tancik, Matthew and Hedman, Peter and Martin-Brualla, Ricardo and Srinivasan, Pratul P.. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)* (2021). `[barron2021mipnerf]`
5. **Mip-NeRF 360** — Barron, Jonathan T. and Mildenhall, Ben and Verbin, Dor and Srinivasan, Pratul P. and Hedman, Peter. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2022). `[barron2022mipnerf360]`
6. **Improving Image Generation with Better Captions** — Betker, James and Goh, Gabriel and Jing, Li and Brooks, Tim and Wang, Jianfeng and Li, Linjie and Ouyang, Long and Zhuang, Juntang and Lee, Joyce and Guo, Yufei and others. (2023). `[betker2023dalle3]`
7. **Low-Complexity Single-Image Super-Resolution Based on Nonnegative Neighbor Embedding** — Bevilacqua, Marco and Roumy, Aline and Guillemot, Christine and Alberi-Morel, Marie-Line. *Proceedings of the British Machine Vision Conference (BMVC)* (2012). `[bevilacqua2012set5]`
8. **ZoeDepth** — Bhat, Shariq Farooq and Birkl, Reiner and Wofk, Diana and Wonka, Peter and Mu. *arXiv preprint arXiv:2302.12288* (2023). `[bhat2023zoedepth]`
9. **Multi-view stereo-regulated NeRF** — Bian, Fenghua and Xiong, Shiyu and Yi, Ran and others. *The Visual Computer* (2025). `[bian2025mvsregnerf]`
10. **Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale** — Bianchi, Federico and Kalluri, Pratyusha and Durmus, Ece and Ladd, Faisal and Choi, Myra and Hashimoto, Tatsunori. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)* (2023). `[bianchi2023diffbias]`
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18. Change Log

Version	Date	Changes
v1.0.0	(submission)	Initial public release: corpus, taxonomy labels, benchmark tables, weakness annotations, severity calculator, deployment thresholds, documentation.

19. Contact and Contributions

Issues, corrections, and additions are welcomed through the repository's issue tracker. Proposed additions to the taxonomy should follow the decision rules (R1–R3) and include a `source_ref`; proposed weakness entries should include the four-dimension schema and per-factor severity scores so the label can be recomputed. Contributions are reviewed for provenance before merging, and each accepted change is recorded in the change log and reflected in the next versioned Zenodo release.

End of Supplementary Material.